RV College of Engineering®, Bengaluru – 59

(An Autonomous Institution affiliated to VTU, Belgavi)

Department of Industrial Engineering and Management

CAD/CAM and Robotics Lab (IM343AI)

SCHEME OF CONDUCT AND EVALUATION

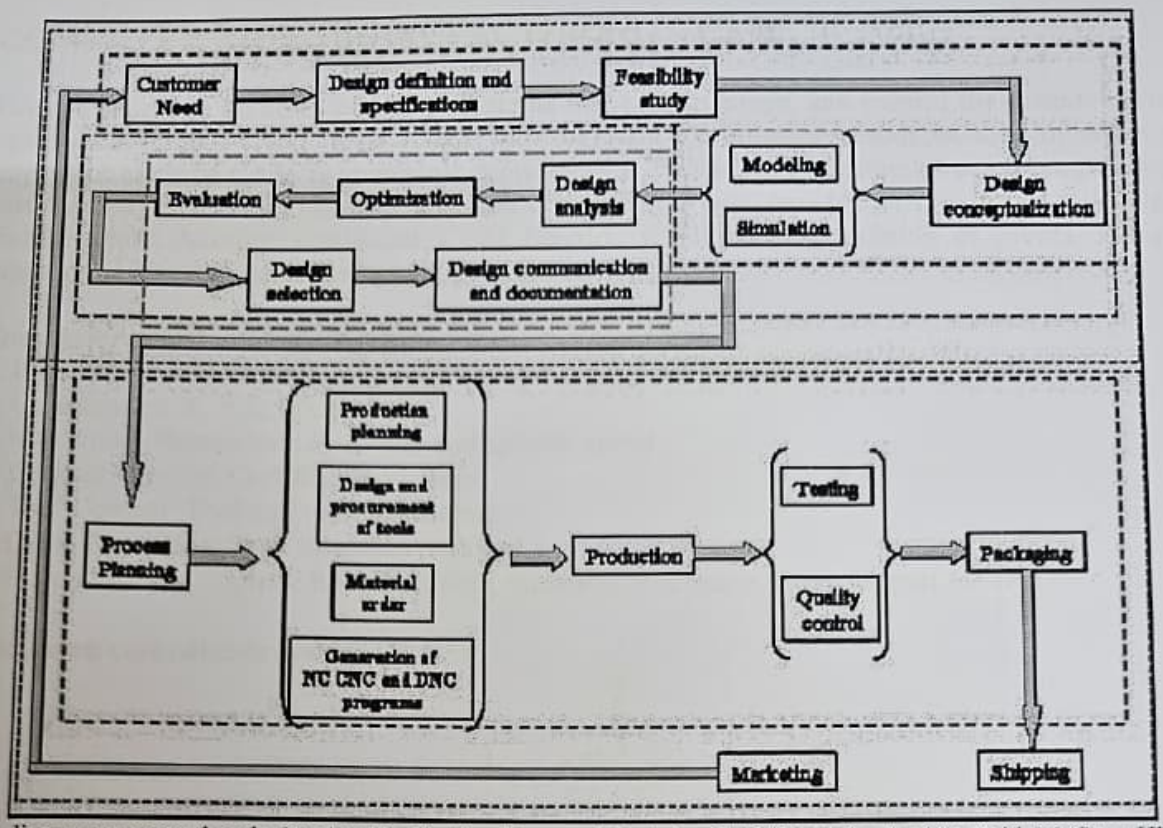
CLASS: IV SEMESTER
YEAR: 2025-26

CIE MARKS: (Max.) : 50
SEE MARKS: (Max) : 50
SEE: 3hrs

SL. No.	Expt. No.	TITLE	No. of Class	Total
		PART – I		
1.	CN01	Programming & Simulation of Turning operation	1	20
2.	CN02	Programming & Simulation of Thread cutting operation	1	20
3.	CN03	Programming & Simulation of Milling operation	1	20
4.	CN04	Programming & Simulation of Mirroring operation	1	20
5.	CN05	Programming & Simulation of Drilling operation	1	20
		PART – II		
6.	RB01	Setting of User Frame and Tool Frame, Jogging the Robot	1	20
7.	RB02	Program to trace a square motion using a Robot using robot.	1	20
8.	RB03	Program to trace a circular motion using a robot using robot.	1	20
9.	RB04	Program to trace a square motion using a Robot(Using Roboguide software)	1	20
10.	RB05	Program to pick and place to workpiece using approach point and retraction point(Using robot or Roboguide software)	1	20
		TEST	1	50
		TOTAL		**250**

EVALUATION SCHEME:

CIE Marks = {Total Class marks (Max.200) + Test marks (Max.50)} Reduced to 50 Marks



A diagram representing the Product Realization Process (Mastering CAD/CAM, by Ibrahim Zeid, McGraw Hill, 2005) is shown above.

Definition of CAD/CAM

Computer Aided Design – CAD

CAD is technology concerned with using computer systems to assist in the creation, modification, analysis, and optimization of a design. Any computer program that embodies computer graphics and an application program facilitating engineering functions in design process can be classified as CAD software.

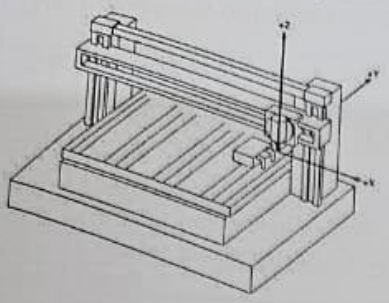
The most basic role of CAD is to define the geometry of design – a mechanical part, a product assembly, an architectural structure, an electronic circuit, a building layout, etc. The greatest benefits of CAD systems are that they can save considerable time and reduce errors caused by otherwise having to redefine the geometry of the design from scratch every time it is needed.

Computer Aided Manufacturing – CAM

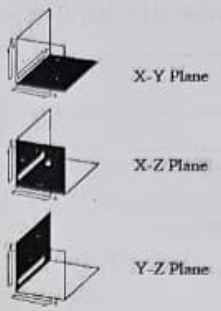
CAM technology involves computer systems that plan, manage, and control the manufacturing operations through computer interface with the plant's production resources.

One of the most important areas of CAM is numerical control (NC). This is the technique of using programmed instructions to control a machine tool, which cuts, mills, grinds, punches or turns raw stock into a finished part. Another significant CAM function is in the programming of robots. Process planning is also a target of computer automation.

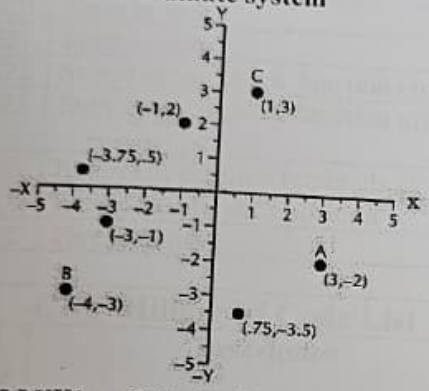
Gantry mill co-ordinate system



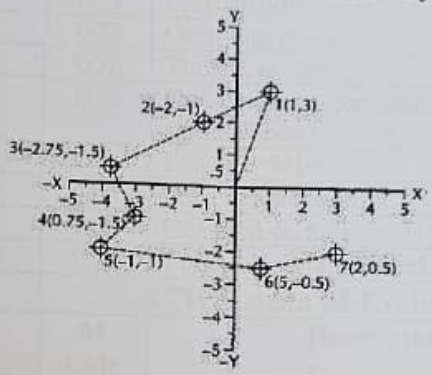
Cartesian co-ordinate system planes



Absolute co-ordinate system



Incremental co-ordinate system



CNC Milling G Code List

G code	Description	G code	Description
G00	Rapid traverse	G59	Workpiece coordinate system 6 selection
G01	Linear interpolation	G68	Coordinate rotation
G02	Circular interpolation CW	G69	Coordinate rotation cancel
G03	Circular interpolation CCW	G73	Peck drilling cycle
G04	Dwell	G74	Left-spiral cutting circle
G17	X Y plane selection	G76	Fine boring cycle
G18	Z X plane selection	G80	Canned cycle cancel
G19	Y Z plane selection	G81	Drilling cycle, spot boring cycle
G28	Return to reference position	G82	Drilling cycle or counter boring cycle
G30	2nd, 3rd & 4th reference position return	G83	Peck drilling cycle
G40	Cutter compensation cancel	G84	Tapping cycle
G41	Cutter compensation left	G85	Boring cycle
G42	Cutter compensation right	G86	Boring cycle
G43	Tool length compensation + direction	G87	Back boring cycle
G44	Tool length compensation – direction	G88	Boring cycle
G49	Tool length compensation cancel	G89	Boring cycle
G53	Machine coordinate system selection	G90	Absolute command
G54	Workpiece coordinate system 1 selection	G91	Increment command
G55	Workpiece coordinate system 2 selection	G92	Setting for work coordinate system or clamp at maximum spindle speed
G56	Workpiece coordinate system 3	G98	Return to initial point in canned cycle

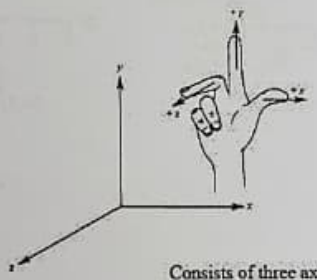
COMPUTER AIDED MANUFACTURING – CAM

CAM technology involves computer systems that plan, manage, and control the manufacturing operations through computer interface with the plant's production resources. One of the most important areas of CAM is numerical control (NC). This is the technique of using programmed instructions to control a machine tool that cuts, mills, grinds, punches or turns raw stock into a finished part. Another significant CAM function is in the programming of robots. Process planning is also a target of computer automation.

Parameters used in CNC Machine.

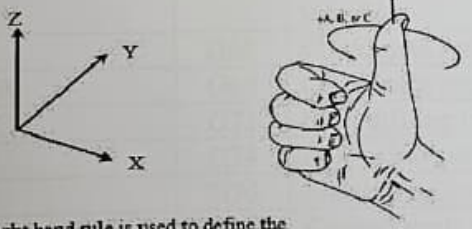
1. Preparatory Information: units, incremental or absolute positioning
2. Coordinates: X, Y,Z, RX,RY,RZ
3. Machining Parameters: Feed rate and spindle speed
4. Coolant Control: On/Off, Flood, Mist
5. Tool Control: Tool and tool parameters
6. Cycle Functions: Type of action required
7. Miscellaneous Control: Spindle on/off, direction of rotation, stops for part movement

Cartesian co-ordinate system



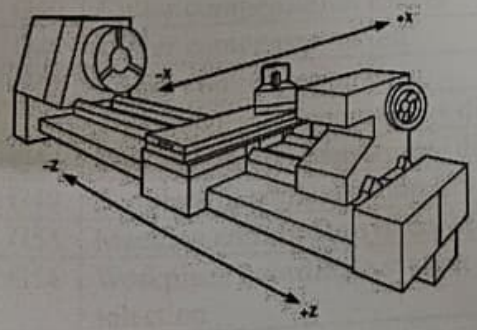
Consists of three axes positioned 90 degrees from each other.

Right hand rule

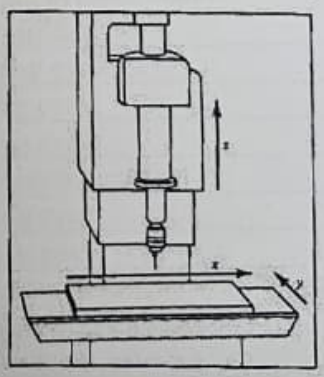


The right hand rule is used to define the positive direction of the coordinate axes.

Standard lathe co-ordinate system



Vertical mill co-ordinate system

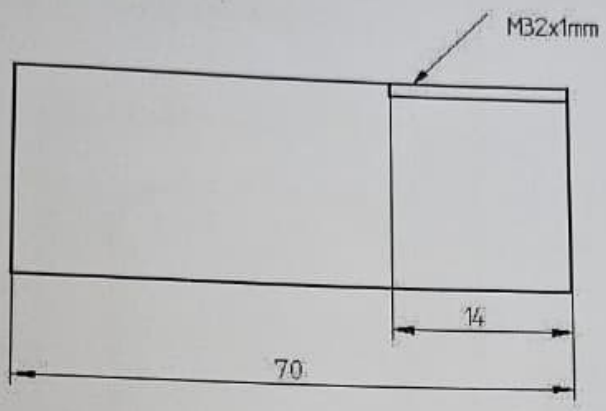


CN02

PROGRAMMING & SIMULATION OF THREAD CUTTING OPERATION

Program: Generate program for threading operation for the profile given below.

Cutting tool: RH turning tool.



Thread Cutting: G76 Cycle

Syntax

G76 P(_A_) (_B_) (_C_) Q R

G76 X Z P Q F

Where,

P(_A_) - No. of Finishing passes

P(_B_) - Thread Chamfer

P(_C_) - Thread Angle

Q - Depth of cut (in micron)

R - Finishing Allowance

X - Minor diameter

Z - Thread length

P - Thread height (in micron)

F - Pitch of the thread

To find minor diameter

$d = D - 2P$

$= 32 - 2(0.613)$

$= 32 - 1.226$

$d = 30.774$

Where

$P = 0.613 * \text{pitch of the thread}$

$P = 0.613 * 1 = 0.613$

Note: d – Minor diameter

D – Major diameter

	selection		
G57	Workpiece coordinate system 4 selection	G99	Return to R point in canned cycle
G58	Workpiece coordinate system 5 selection		

CNC Lathe G Code List

G00	Rapid traverse	G40	Tool nose radius compensation cancel
G01	Linear interpolation	G41	Tool nose radius compensation left
G02	Circular interpolation CW	G42	Tool nose radius compensation right
G03	Circular interpolation CCW	G70	Finish machining cycle
G04	Dwell	G71	Turning cycle
G09	Exact stop	G72	Facing cycle
G10	Programmable data input	G73	Pattern repeating cycle
G20	Input in inch	G74	Peck drilling cycle
G21	Input in mm	G75	Grooving cycle
G22	Stored stroke check function on	G76	Threading cycle
G23	Stored stroke check function off	G92	Coordinate system setting or max. spindle speed setting
G27	Reference position return check	G94	Feed Per Minute
G28	Return to reference position	G95	Feed Per Revolution
G32	Thread cutting	G96	Constant surface speed control
		G97	Constant surface speed control cancel

CNC Milling M Code List

M code	Description
M00	Program stop
M01	Optional program stop
M02	End of program
M03	Spindle start forward CW
M04	Spindle start reverse CCW
M05	Spindle stop
M06	Too change
M07	Coolant ON – Mist coolant/Coolant thru spindle
M08	Coolant ON – Flood coolant
M09	Coolant OFF
M19	Spindle orientation
M28	Return to origin
M29	Rigid tap
M30	End of program (Reset)
M41	Low gear select
M42	High gear select
M94	Cancel mirrorimage
M95	Mirrorimage of X axis
M96	Mirrorimage of Y axis
M98	Subprogram call
M99	End of subprogram

CNC Lathe M Code List

M code	Description
M00	Program stop
M01	Optional program stop
M02	End of program
M03	Spindle start forward CW
M04	Spindle start reverse CCW
M05	Spindle stop
M08	Coolant on
M09	Coolant off
M29	Rigid tap mode
M30	End of program reset
M40	Spindle gear at middle
M41	Low Gear Select
M42	High Gear Select
M68	Hydraulic chuck close
M69	Hydraulic chuck open
M78	Tailstock advancing
M79	Tailstock reversing
M94	Mirrorimage cancel
M95	Mirrorimage of X axis
M98	Subprogram call
M99	End of subprogram

RB01

ROBOT PROGRAMMING

INTRODUCTION

Robots are devices that are programmed to move parts, or to do work with a tool. Robotics is a multidisciplinary engineering field dedicated to the development of autonomous devices, including manipulators and mobile vehicles.

An industrial robot is a general – purpose, programmable machine possessing certain anthropomorphic characteristics. The most typical anthropomorphic or human like, characteristics of a robot is its arm. This arm, together with the robots capacity to be programmed, make it ideally suited to a variety of production tasks, including machine loading, spot welding, spray painting and assembly. The robot can be programmed to perform sequence of mechanical motions, and it can repeat that motion sequence over the over until programmed to perform some other job.

An industrial robot is a general purpose programmable machine that possesses certain anthropomorphic features

- The most apparent anthropomorphic feature of an industrial robot is its mechanical arm, or manipulator
- Robots can perform a variety of tasks such as loading and unloading machine tools, spot welding automobile bodies, and spray painting
- Robots are typically used as substitutes for human workers in these tasks

An industrial robot is a programmable, multi-functional manipulator designed to move materials, parts, tools, or special devices through variable programmed motions for the performance of a variety of tasks.

An industrial robot consists of a mechanical manipulator and a controller to move it and perform other related functions

- The mechanical manipulator consists of joints and links to position and orient the end of the manipulator relative to its base
- The controller operates the joints in a coordinated fashion to execute a programmed work cycle
- A robot joint is similar to a human body joint It provides relative movement between two parts of the body
- Typical industrial robots have five or six joints, Manipulator joints: classified as linear or rotating

Roobot Physical Configuration

Industrial robots come in a variety of shapes and sizes. They are capable of various arm manipulations and they possess different motion systems.

Classification based on Physical configurations

Four basic configurations are identified with most of the commercially available industrial robots

1. **Cartesian configuration:** A robot which is constructed around this configuration consists of three orthogonal slides, as shown in fig. the three slides are parallel to the x, y, and z axes of the Cartesian coordinate system. By appropriate movements of these slides, the robot is capable of moving its arm at any point within its three dimensional rectangularly spaced work space.
2. **Cylindrical configuration:** in this configuration, the robot body is a vertical column that swivels about a vertical axis. The arm consists of several orthogonal slides which allow the

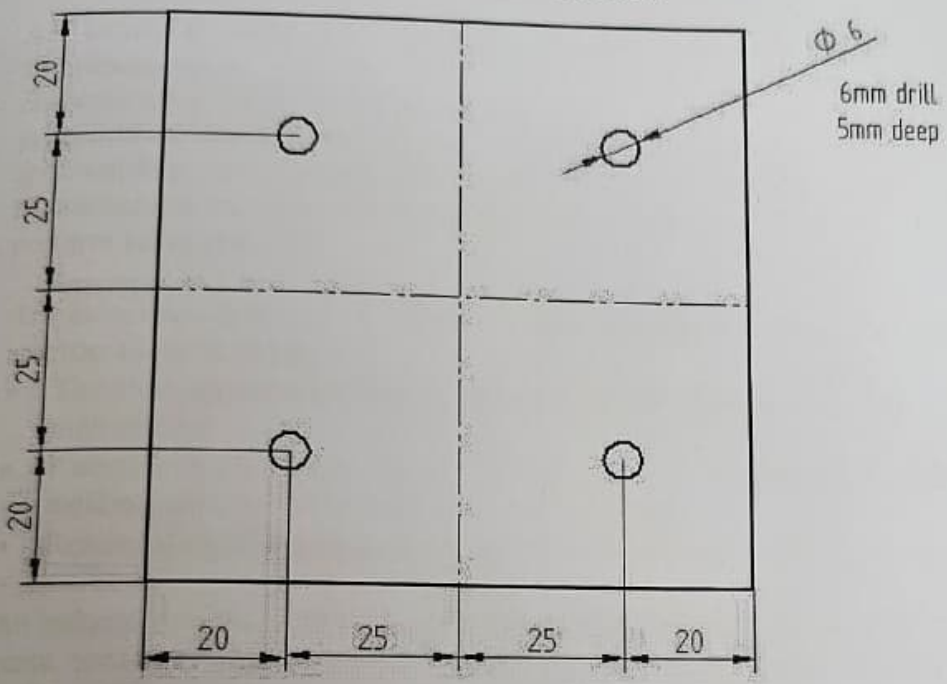
CN05

PROGRAMMING & SIMULATION OF DRILLING OPERATION

Generate a CNC program for performing drilling operation and simulate for the profile given below.

Hole diameter 6mm and depth 5mm.

Cutting tool : Slot drill cutter of 6mm diameter.



G73 X Y Z Q R F K

X,Y – COORDINATE POINT OF DRILLING

Z – DEPTH OF DRILL HOLE

R – RETURN TO REFERENCE POINT

F- FEED RATE

K-NUMBER OF DRILL HOLES

Program

insertion, spot welding, hole drilling, machine loading and unloading, and crude assembly operations.

2. **Continuous-path (CP) control robot:** with CP control, the robot can stop at any specified point along the controlled path. All the points along the path must be stored explicitly in the robot's control memory. Typical applications include spray painting, finishing, gluing, and arc welding operations.
3. **Controlled-path robot:** the control equipment can generate paths of different geometry such as straight lines, circles, and interpolated curves with a high degree of accuracy. All controlled-path robots have a servo capability to correct their path.

Technical Features of an Industrial Robot

The technical features of an industrial robot determine its efficiency and effectiveness at performing a given task. The following are some of the most important among these technical features.

Degree of Freedom (D.O.F) - Each joint on the robot introduces a degree of freedom. Each dof can be a slider, rotary, or other type of actuator. Robots typically have 5 or 6 degrees of freedom. 3 of the degrees of freedom allow positioning in 3D space, while the other 2or 3 are used for orientation of the end effector. 6 degrees of freedom are enough to allow the robot to reach all positions and orientations in 3D space. 5 D.O.F requires a restriction to 2D space, or else it limits orientations. 5 D.O.F robots are commonly used for handling tools such as arc welders.

Work Volume/Workspace - The robot tends to have a fixed and limited geometry. The work envelope is the boundary of positions in space that the robot can reach. For a Cartesian robot (like an overhead crane) the workspace might be a square, for more sophisticated robots the workspace might be a shape that looks like a 'clump of intersecting bubbles'.

Precision Movement

The precision with which the robot can move the end of its wrist is a critical consideration in most applications. In robotics, precision of movement is a complex issue, and we will describe it as consisting of three attributes:

1. Control resolution
2. Accuracy
3. Repeatability

Control Resolution - This is the smallest change that can be measured by the feedback sensors, or caused by the actuators, whichever is larger. If a rotary joint has an encoder that measures every 0.01 degree of rotation, and a direct drive servo motor is used to drive the joint, with a resolution of 0.5 degrees, then the control resolution is about 0.5 degrees (the worst case can be 0.5+0.01).

Accuracy - This is determined by the resolution of the workspace. If the robot is commanded to travel to a point in space, it will often be off by some amount, the maximum distance should be considered the accuracy.

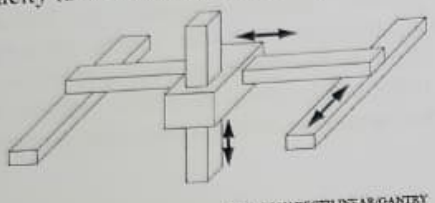
Repeatability - The robot mechanism will have some natural variance in it. This means that when the robot is repeatedly instructed to return to the same point, it will not always stop at the same position.

Speed - refers either to the maximum velocity that is achievable by the TCP, or by individual joints. This number is not accurate in most robots, and will vary over the workspace as the geometry of the robot changes.

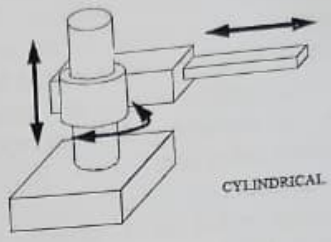
Weight Carrying Capacity (Payload) - The payload indicates the maximum mass the robot can lift before either failure of the robots, or dramatic loss of accuracy. It is possible to exceed the maximum payload, and still have the robot operate, but this is not advised. When the robot is

arm to be moved up or down and in and out with respect to the body. This is illustrated schematically in figure.

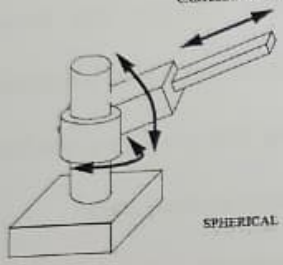
3. **Polar configuration:** this configuration also goes by the name "spherical coordinate" because the workspace within which it can move its arm is a partial sphere as shown in figure. The robot has a rotary base and a pivot that can be used to raise and lower a telescoping arm.
4. **Jointed-arm configuration:** is combination of cylindrical and articulated configurations. This is similar in appearance to the human arm, as shown in fig. the arm consists of several straight members connected by joints which are analogous to the human shoulder, elbow, and wrist. The robot arm is mounted to a base which can be rotated to provide the robot with the capacity to work within a quasi-spherical space.



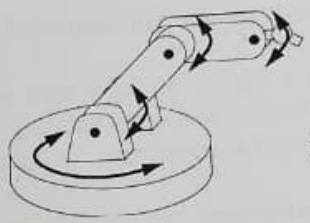
CARTESIAN/RECTILINEAR/GANTRY



CYLINDRICAL



SPHERICAL



ARTICULATED/REVOLUTE/ JOINTED SPHERICAL

Basic Robot Motions

Whatever the configuration, the purpose of the robot is to perform a useful task. To accomplish the task, an end effector, or hand, is attached to the end of the robots arm. It is the end effector which adapts the general purpose robot to a particular task. To do the task, the robot arm must be capable of moving the end effectors through a sequence of motions and positions.

Six degrees of freedom

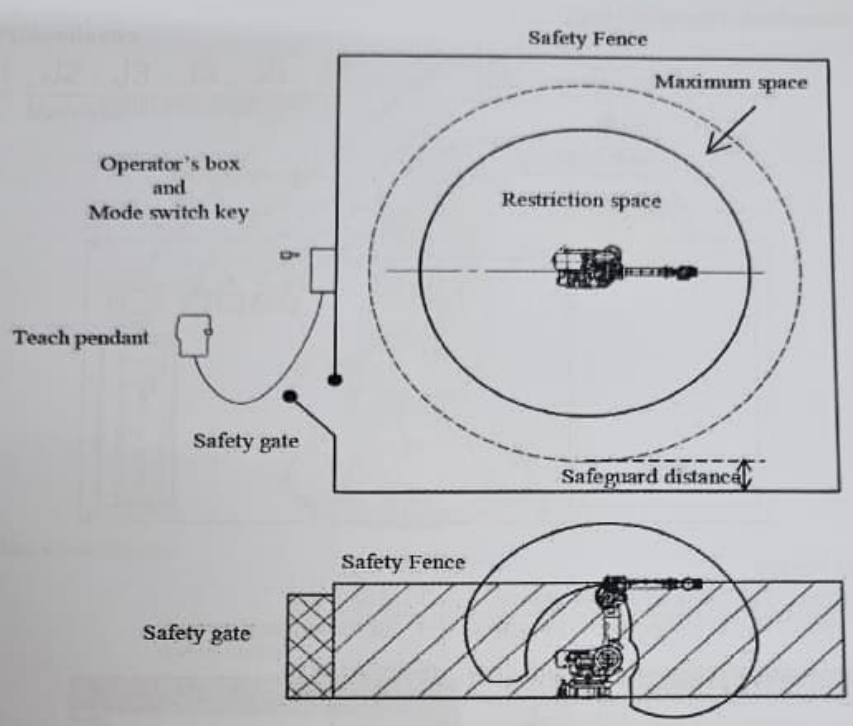
There are six basic motions or degrees of freedom, which provide the robot with the capability to move the end effectors through the required sequences of motions. These six degree of freedom are intended to emulate the versatility of movement possessed by the human arm. Not all robots are equipped with the ability to move in all sex degrees. The six basic motions consist of three arm and body motions and three wrist motions.

Arm and body motions

1. Vertical traverse: Up and down motion of the arm, caused by pivoting the entire arm about a horizontal axis or moving the arm along a vertical slide.
2. Radial traverse: extension and retraction of the arm (in and out movement)
3. Rotational traverse: rotation about the vertical axis (right or left swivel of the robot arm) WRIST MOTION
4. Wrist swivel: Rotation of the wrist
5. Wrist bend: Up or down movement of the wrist, this also involves rotation movement.
6. Wrist yaw: Right or left swivel of the wrist.

Motion system:

1. **Point-to-point (PTP) control robot:** is capable of moving from one point to another point. The locations are recorded in the control memory. PTP robots do not control the path to get from one point to the next point. Common applications include component



Setting of User Frame and Tool Frame

Jogging the Robot

- Release E-Stops
- Enable TP & Change the Mode to T1
- Select Joint by pressing key "coord"
- Select low feed rate
- Hold dead man switch
- Change to position screen
- Hold Shift + J1 or J2……

accelerating fast, the payload should be less than the maximum mass. This is affected by the ability to firmly grip the part, as well as the robot structure, and the actuators. The end of arm tooling should be considered part of the payload.

Type of Drive System

There are three basic drive system used in commercially available robots:

1. **Hydraulic drive:** gives a robot great speed and strength. These systems can be designed to actuate linear or rotational joints. The main disadvantage of a hydraulic system is that it occupies floor space in addition to that required by the robot.
2. **Electric drive:** compared with a hydraulic system, an electric system provides a robot with less speed and strength. Accordingly, electric drive systems are adopted for smaller robots. However, robots supported by electric drive systems are more accurate, exhibit better repeatability, and are cleaner to use.
3. **Pneumatic drive:** are generally used for smaller robots. These robots, with fewer degrees of freedom, carry out simple pick-and-place material handling operations.

Safety Precautions for the Operator

- Install safety fence
- Do not go inside the robot fence
- Provide interlock for fence door
- Emergency stop buttons must be with in operators reach
- Emergency stop buttons should be pressed when the robot is not in operation

Safety Precautions for the Programmer

- Perform teaching by T1 Mode
- Check the program by T1/T2 modes with Step
- No one should enter into fence during teaching
- Remove the key from the SOP.

Operating Modes

- T1: Test Mode 1
  - Program can be activated from the teach pendant only
  - The ROBOT cannot be operated more than 250mm/sec
  - Safety Fence is disabled
- T2: Test Mode 2
  - Program can be activated from the teach pendant only
  - The ROBOT can be operated at the specified maximum speed.
  - Safety Fence is disabled
- AUTO: Automatic Mode
  - The Operator or Panel box becomes enable
  - The ROBOT program can be called by the operator from the operator panel itself by pressing the cycle start or Peripheral I/O.
  - Safety fence is enabled
  - The ROBOT can be operated at the specified maximum speed

Don'ts

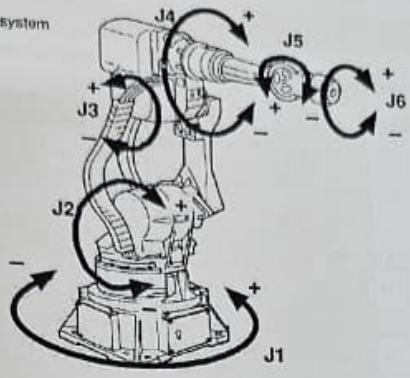
- Do not walk on Teach Pendent cable.
- Do not open the cabinet door unnecessarily.
- Do not use force air on PCBs.
- Do not remove the Mechanical unit battery in Power off condition.
- Do not remove the eye mark on each joints.
- Do not step-on at any part of Robot.
- Do not plug/unplug cables or fuses in power on condition

Joint Coordinates

(J1 , J2 , J3 , J4 , J5 , J6)

J1, J2, J3: Main axis; J4, J5, J6: Wrist axis

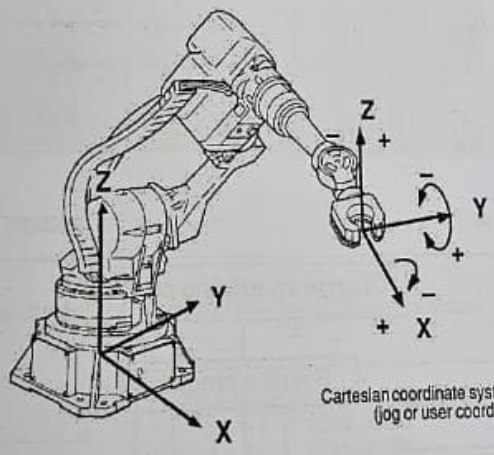
Joint coordinate system



Cartesian Coordinates

UF , UT , (X , Y , Z , W , P , R)

UF: User coordinate system number; UT: Tool coordinate system number; (X, Y, Z): Position; (W, P, R): Attitude



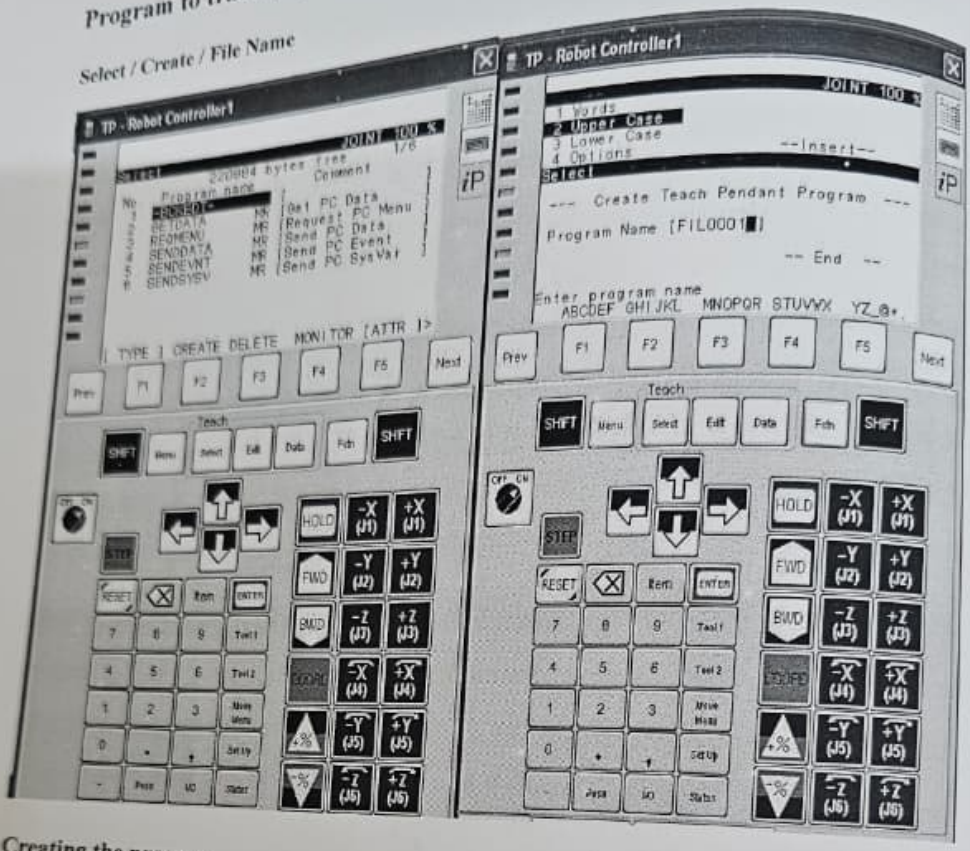
Cartesian coordinate systems (jog or user coordinate systems)

Inference:

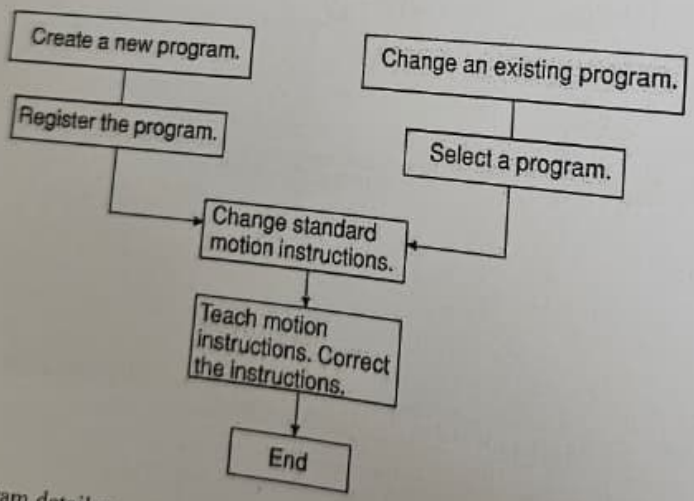
Particulars	Marks awarded	Marks obtained
1. Execution	10	
2. Inference	5	
3. Viva	5	
Total	20	
Signature with Date		

Program to trace a square motion using a Robot

Select / Create / File Name

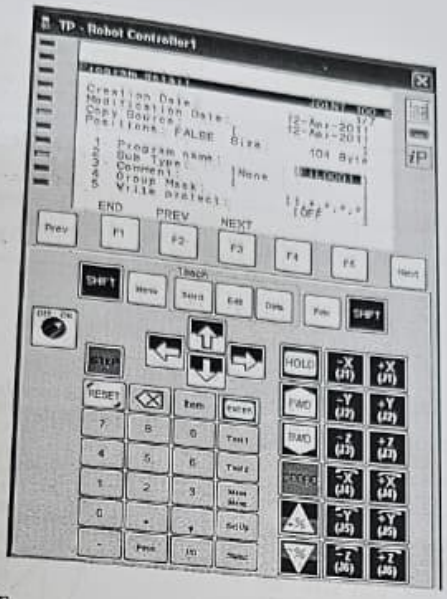


Creating the program



Program detail screen, provides information such as created date, size, write protection comments etc.

RB02



Teaching a motion instruction
Move the robot to the position in the work area by jog feed
Move the cursor to the end

```
SAMPLE1                                  JOINT 30%
[End]                                        1/1


POINT                                    TOUCHUP >
```

Press F1[POINT] key to display the standard motion instruction menu

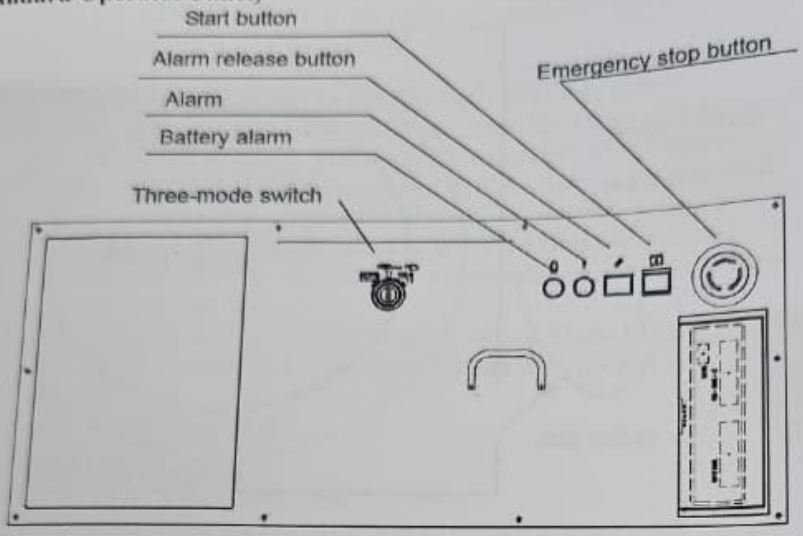
```
POINT
```

F1

```
Joint default menu                       JOINT 30%
 1  J  P[ ] 100% FINE
 2  J  P[ ] 100% FINE
 3  L  P[ ] 1000cm/min CNT50
 4  L  P[ ] 1000cm/min CNT50
SAMPLE3
                                             1/1
[End]

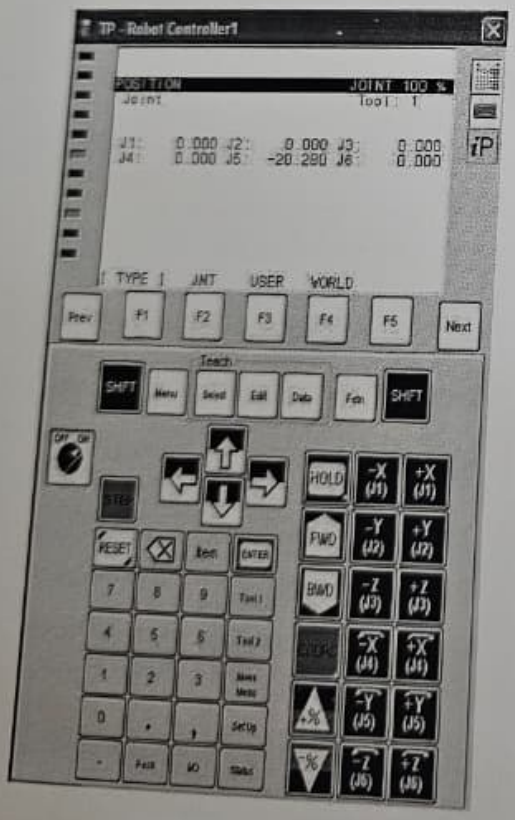
ED_DEF                                   TOUCHUP >
```

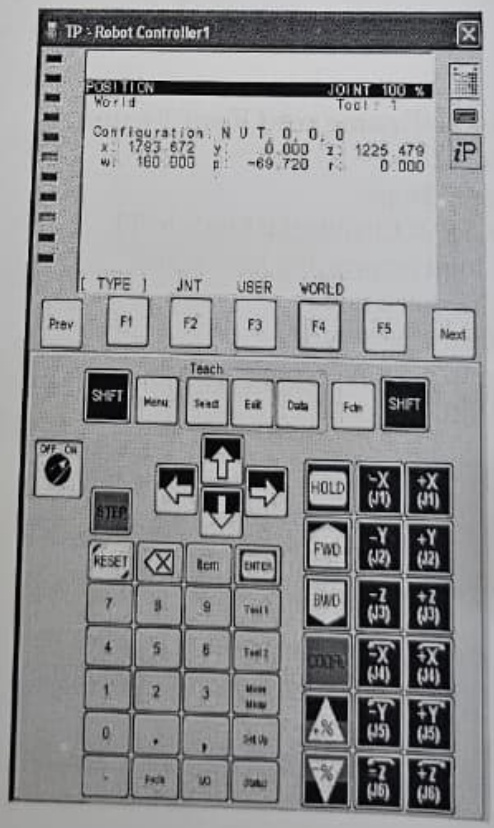
SOP (Standard Operator Panel)

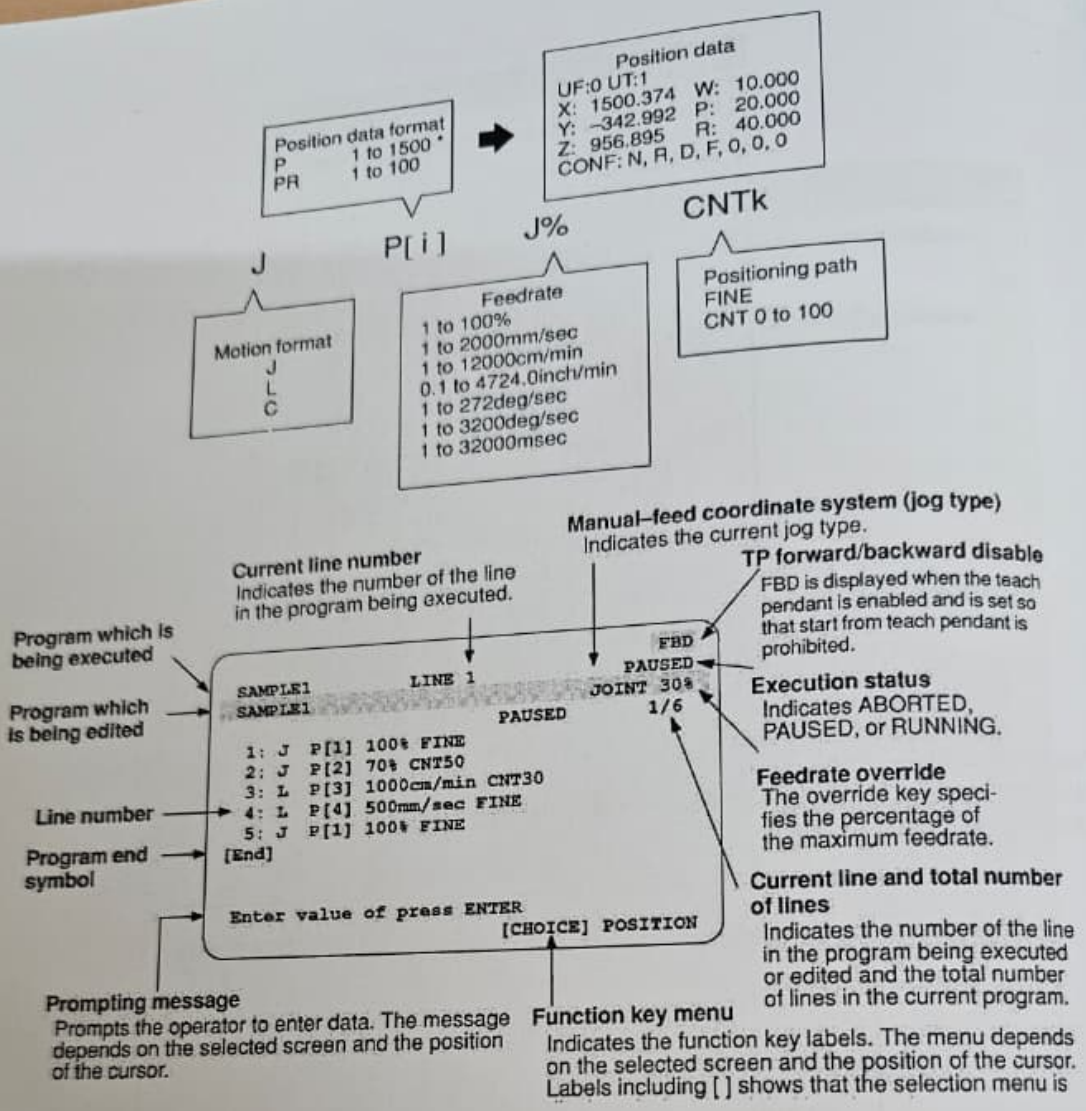


Position Screen

- Press "Pos" to get position screen
- Press "F2" to joint coordinates, press "F4" for world coordinates

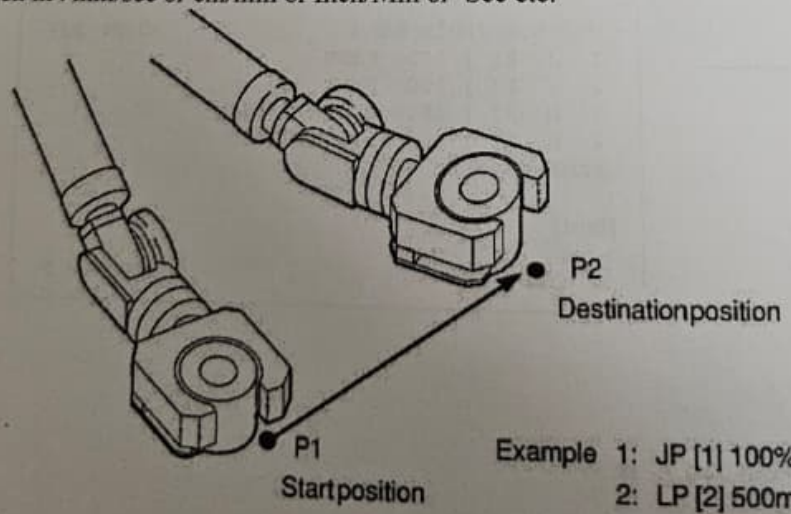






L Linear motion :

Tool tip will move linearly towards the destination.
All the joints are coordinates so that straight line motion is achieved.
Speed is given in : mm/sec or cm/min or Inch/Min or Sec etc.



Example 1: JP [1] 100% FINE
2: LP [2] 500mm/sec FINE

RB04

Program to pick and place to workpiece using approach point and Retraction Point

Select / Create / File Name

Press F1[POINT] key to display the standard motion instruction menu

1 J P [1] 100% FINE
2 L P [2] 100 mm/sec FINE
3 L P [3] 100 mm/sec FINE
4 RO [1: ON : GRIPPER OPEN] = OFF(For the controller, telling to it should be off for safety)
5 RO [2: OFF : GRIPPER CLOSE] = ON(holding the job firmly)
6 J P [2] 100% FINE
7 J P [4] 100% FINE
8 J P [5] 100% FINE
9 RO [2: OFF : GRIPPER CLOSE] = ON (For the controller, telling to it should be on)
10 RO [1: ON : GRIPPER OPEN] = OFF (releasing the job)
11 J P [4] 100% FINE
12 J P [1] 100% FINE

Program

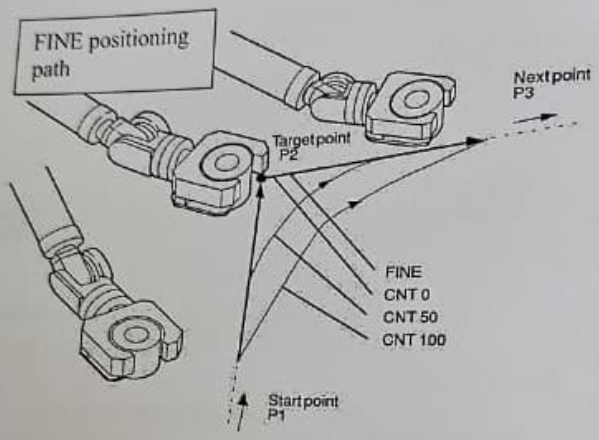
Particulars	Marks awarded	Marks obt
1. Write-up of program	10	
2. Execution	5	
3. Results and Inference	2	
4. Viva	3	
Total	20	

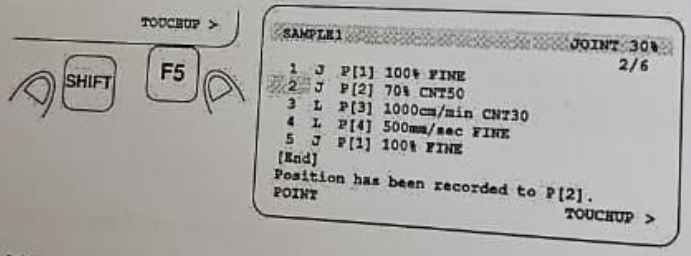
RB03

Program to trace a circular motion using a Robot

Steps

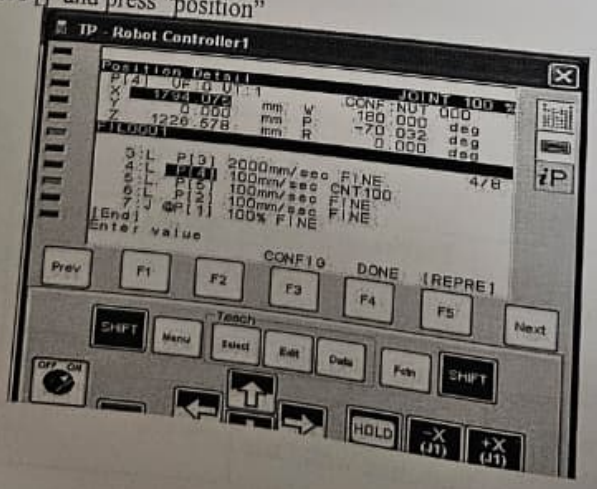
- Ensure that passing point is somewhere in the middle of the arc
- Teaching should be as good as possible
- The arc need to be less than or equal to semi circle
- If full circle need to be made, divide it into 2 or more pieces
- Orientation need to be changed if required
- Move robot to start point of arc and record P1
- Move robot to end point and record it by "Touchup"





Program steps

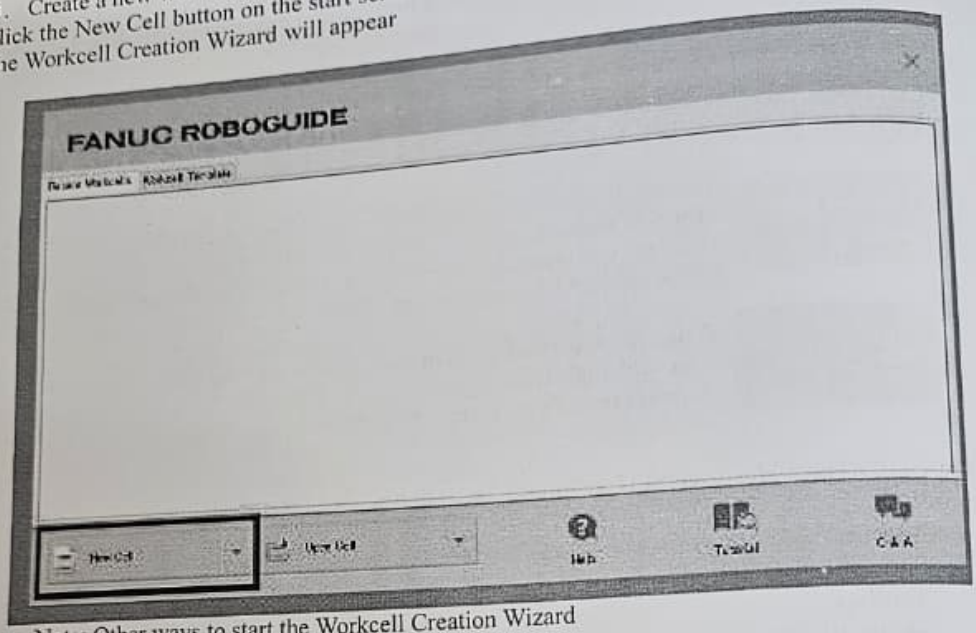
- Recorded position can be viewed
- Place cursor on P[] and press "position"



Workcell layout tutorial

1. Create a new workcell

Click the New Cell button on the start screen.
The Workcell Creation Wizard will appear

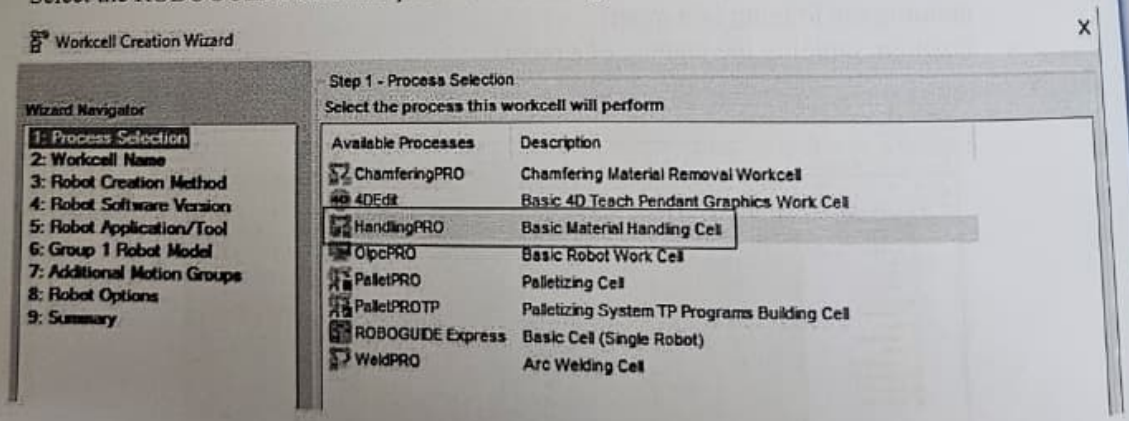


Note: Other ways to start the Workcell Creation Wizard

You can also start the Workcell Creation Wizard in the following operations.

1. On the File menu, select New Cell.
2. Click the New Cell icon on the toolbar.

2. Select the process from the list.
   Select the ROBOGUIDE standard process "HandlingPRO" and click the Next button.



3. Enter a name for the workcell.
   Enter "SimProg1" and click the Next button.

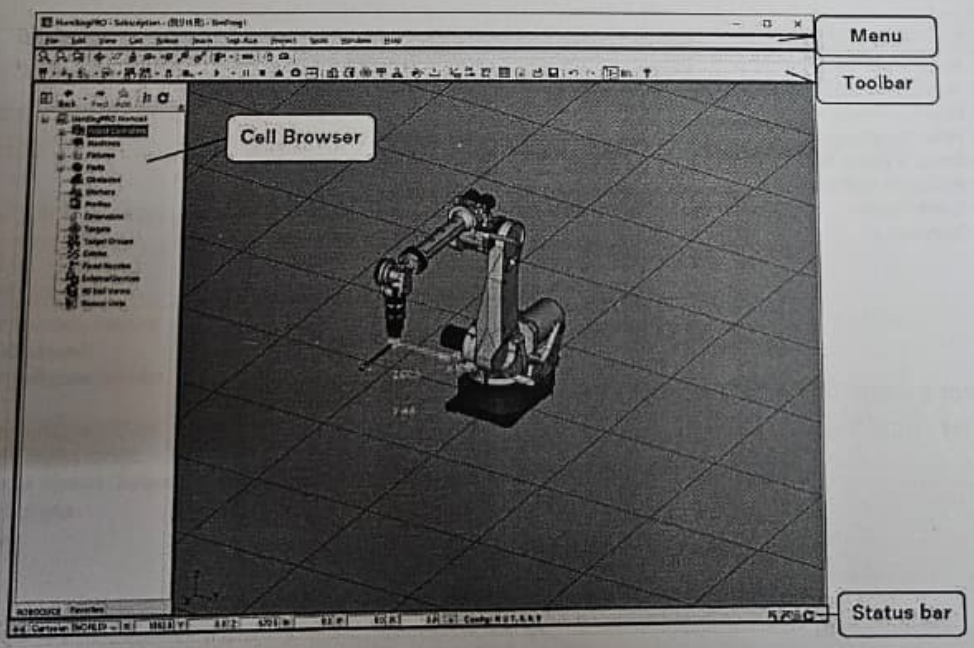
Roboguide Demonstration

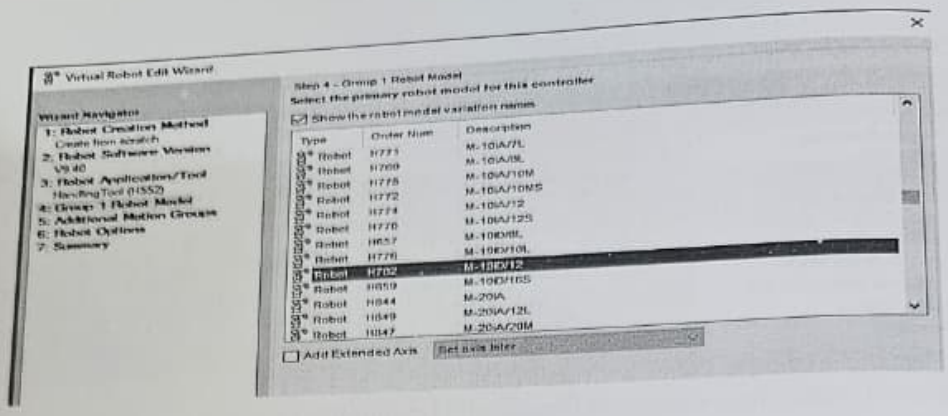
ROBOGUIDE is a software that supports each step from design process to operation check for a robot system.

Layout a workcell	•You can place objects in the workcell and modify a layout by mouse operation or numerical input. ROBOGUIDE provides many CAD data of fixtures, parts, and so on.
Create a robot program	•You can create a robot program for use in a real environment by using a variety of functions that support program creation. There are two methods to create robot programs: Simulation Program Editor and Virtual Teach Pendant Editor.
Check a robot motion	•You can execute the program you have created and estimate the cycle time with high accuracy. You can transfer the created program to an actual robot.

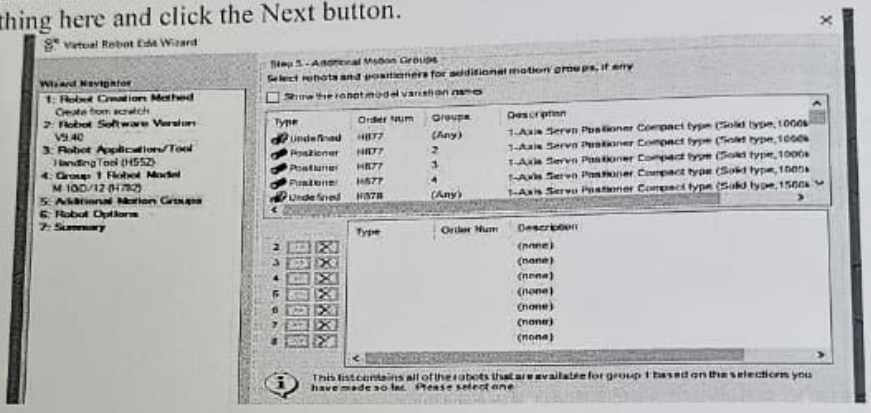
ROBOGUIDE provides several User Interfaces so that you can use them according to the purpose.

- **Menu:**
  Shows ROBOGUIDE operation menus.
- **Toolbar:**
  Shows the shortcut icons for the basic menus.
  The toolbar lets you easily access the menu.
- **Cell Browser:**
  Shows all elements constituting a workcell in the tree structure.
  The Cell browser lets you quickly find out the elements on workcell.
- **Status bar:**
- Shows a current position of the robot and operation icons when clicking a tool of a robot. The status bar lets you easily perform operations (such as modifying a robot position, undoing the jogging of a robot).
- Position, undoing the jogging of a robot).

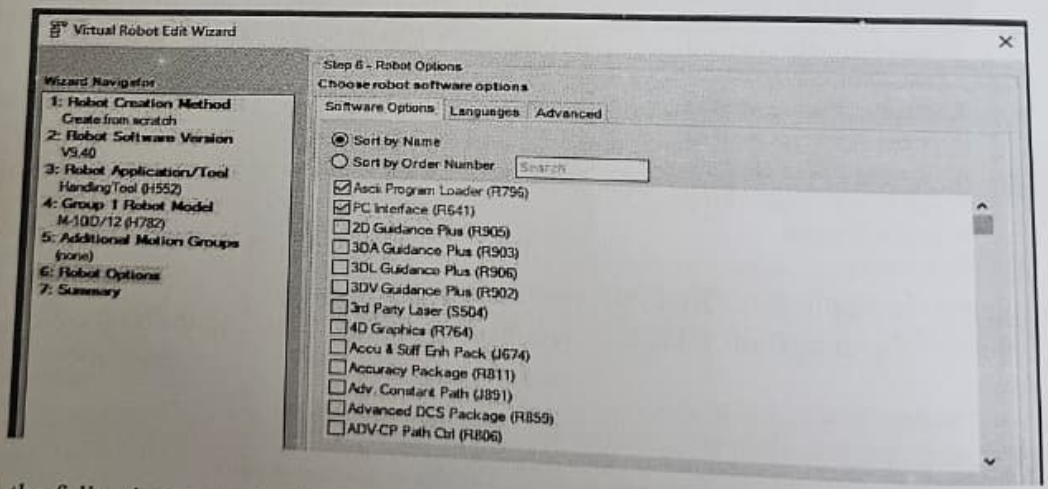




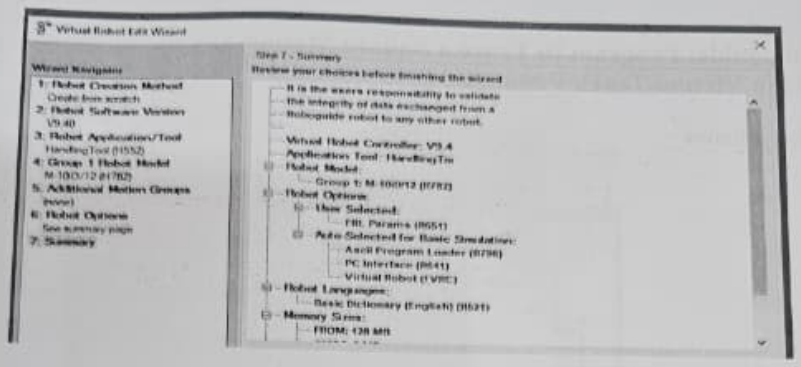
8. Set the Additional Motion Groups.
   Select nothing here and click the Next button.



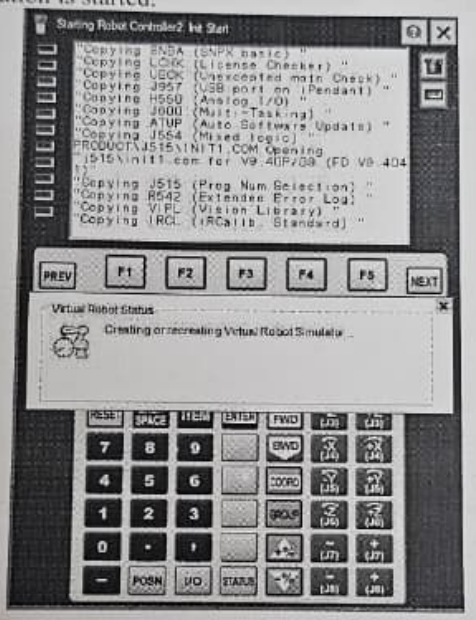
9. Select robot options you need.
   Select nothing here and click the Next button.



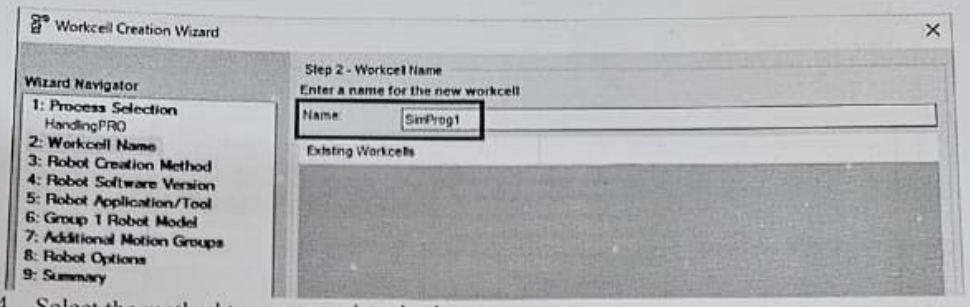
10. Check the following contents displayed in the Summary screen.
    1. **Process** : HandlingPRO
    2. **Workcell Name** : SimProg1
    3. **Virtual Robot Controller** : "V9.40"
    4. **Application Tool** : HandlingTool
    5. **Robot Model** : R-2000iC/165F (H721)



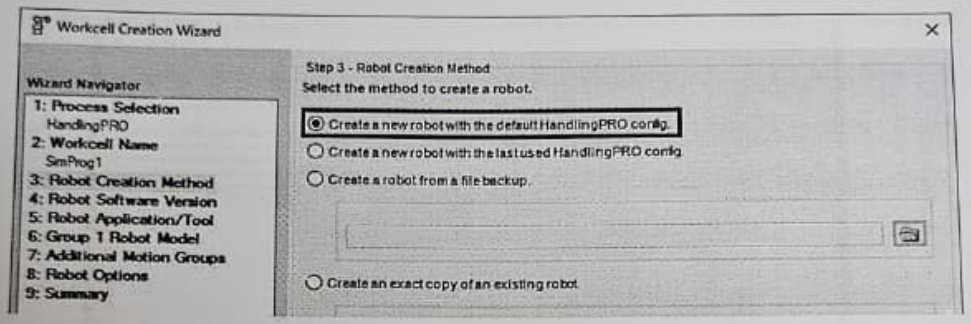
11. Click the Finish button.
    The virtual robot creation is started.



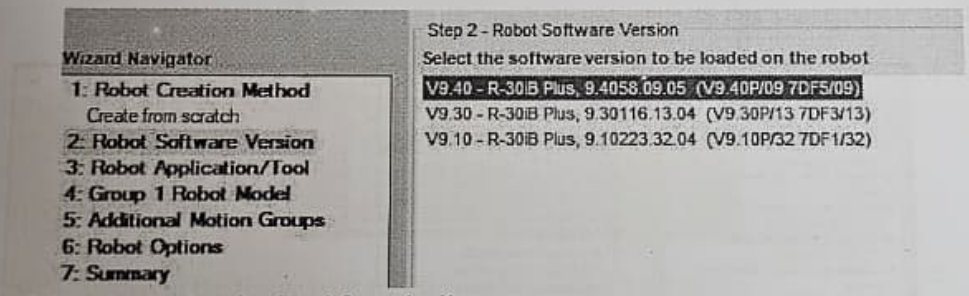
Click "1" button in the flange type selection and click the Enter button.



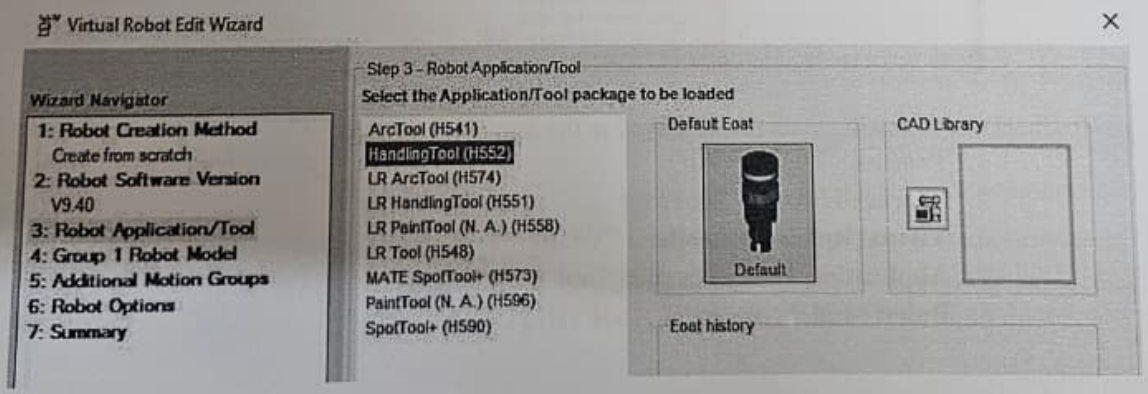
4. Select the method to create a virtual robot.
   Select "Create a new robot with the default HandlingPRO config." and click the Next button.



5. Select the software version from the list.
   Select the V9.40 software version and click the Next button.



6. Select the Application/Tool from the list.
   Select "HandlingTool（H552）" and the Default Eoat, and click the Next button.

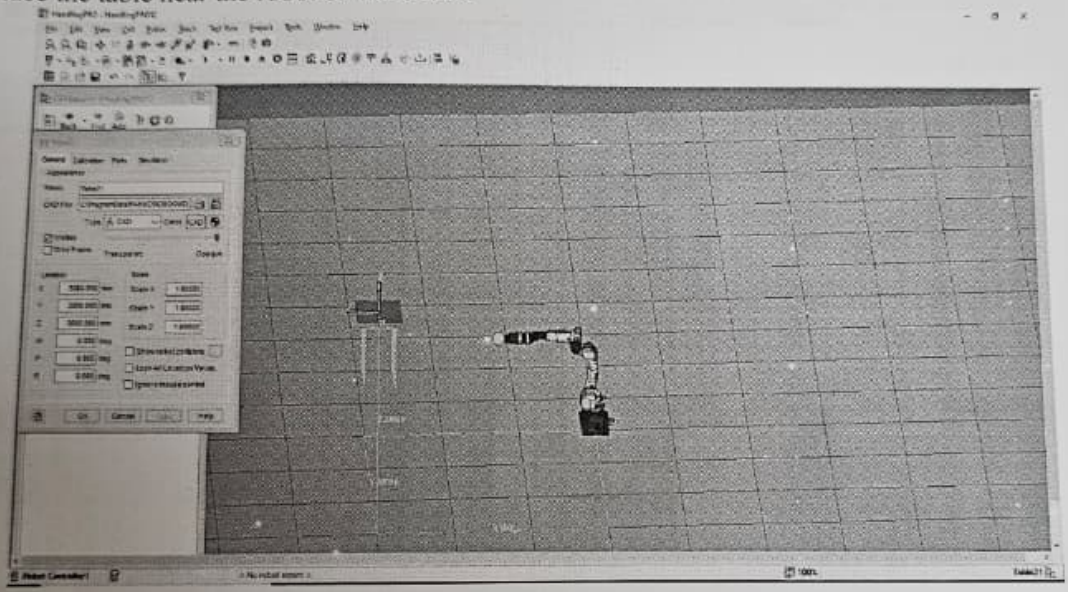


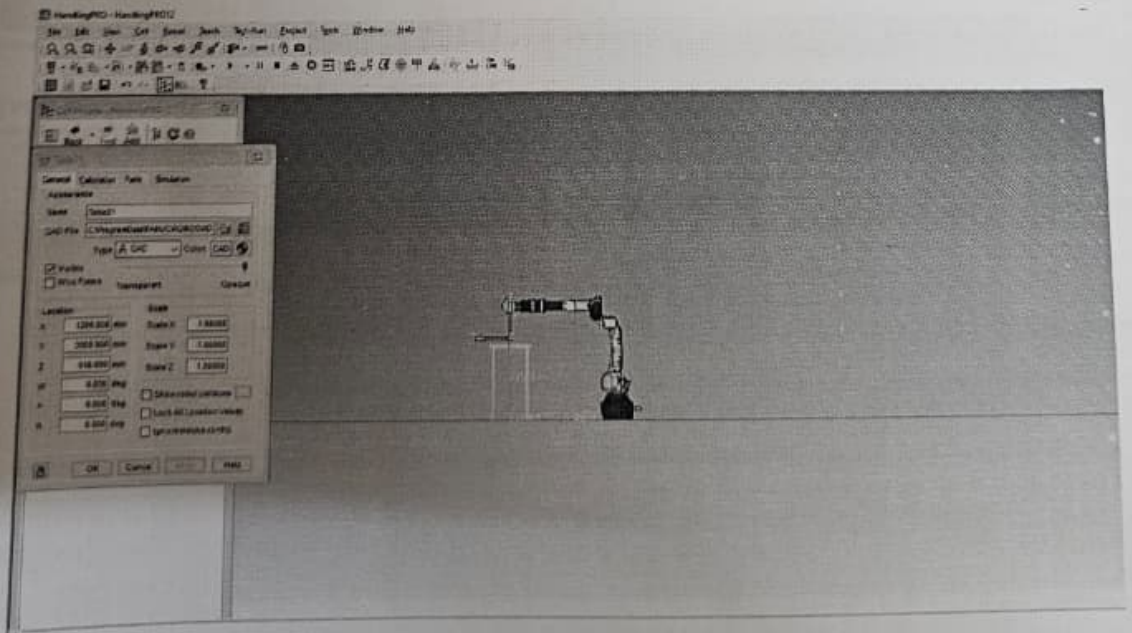
7. Select the robot model from the list.
   Select "M-10iD/12" and click the Next button.

Select table 21



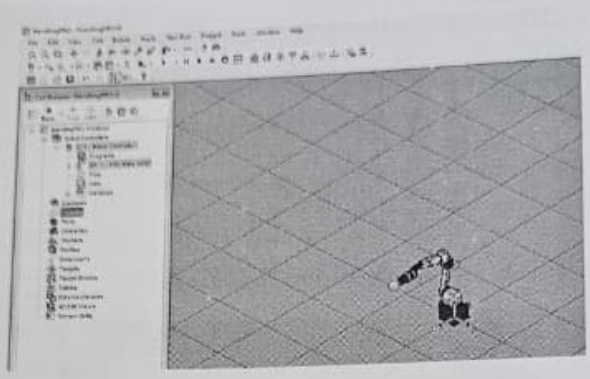
Place the table near the robot on the screen



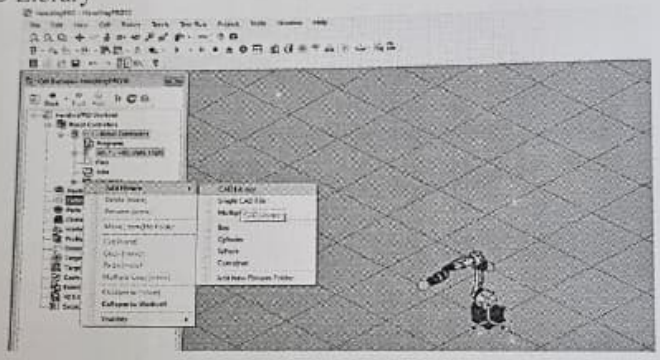


Roboguide: Program to Trace a Square Motion Using Virtual Teach Pendent.

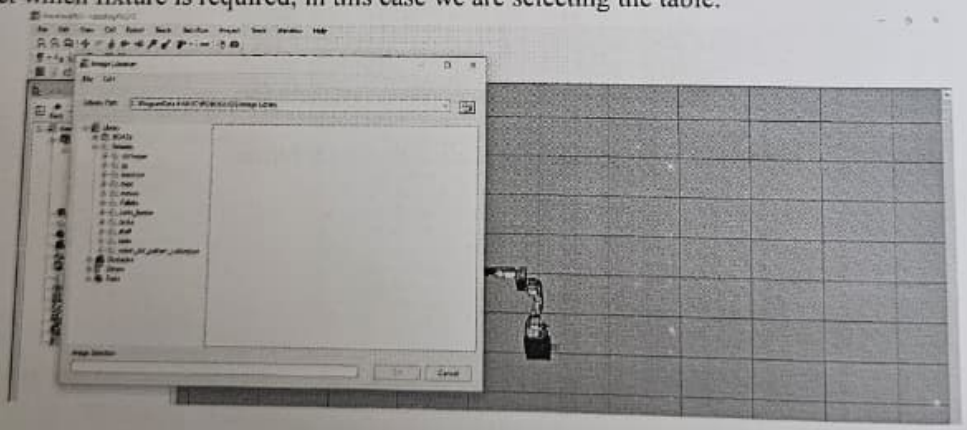
Select a fixture



Add fixture, CAD Library



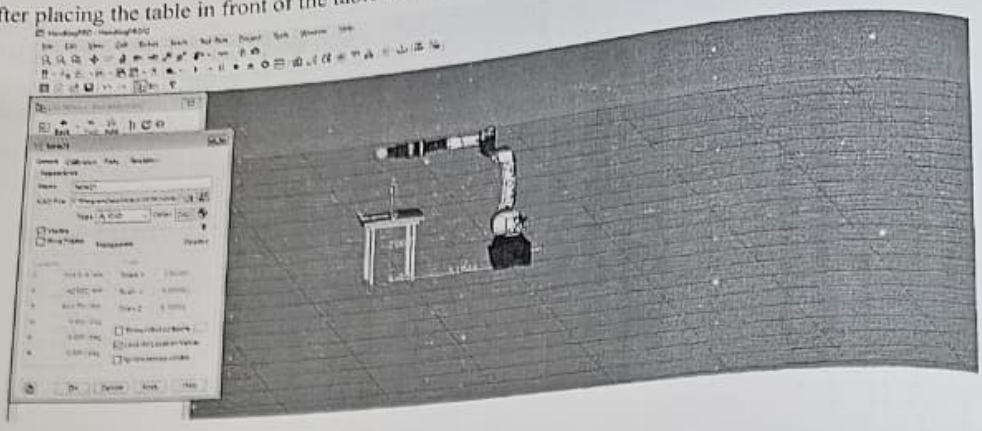
Select which fixture is required, in this case we are selecting the table.

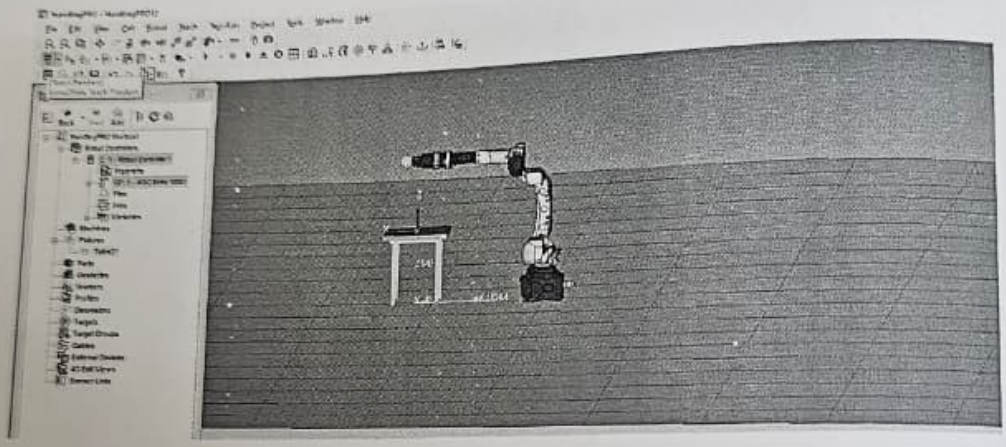




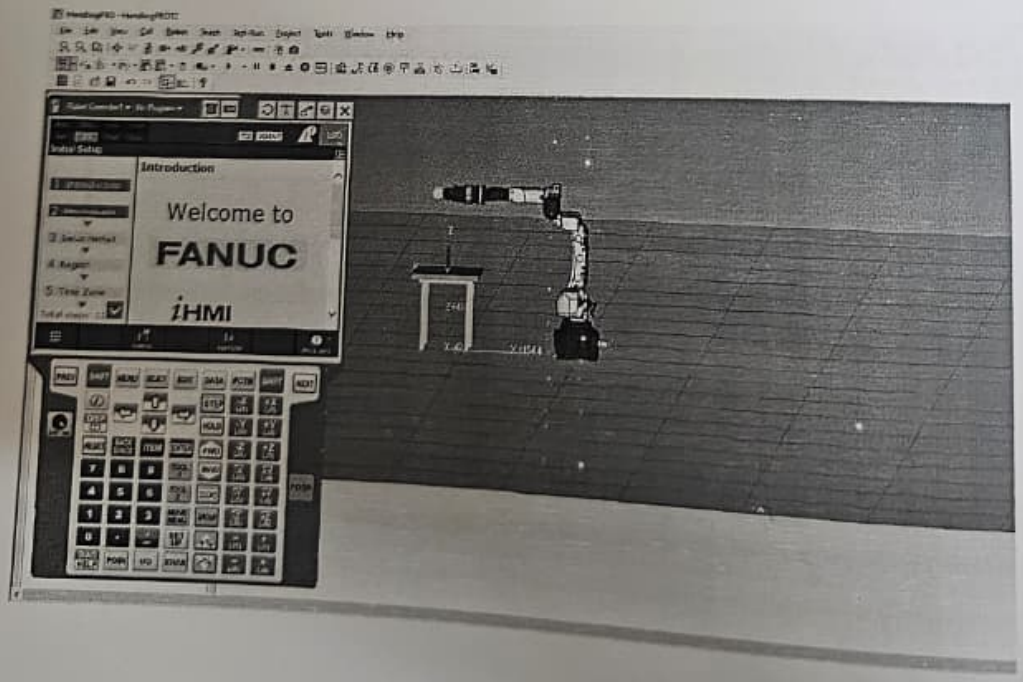


After placing the table in front of the table lock all the dimensions of both robot and table

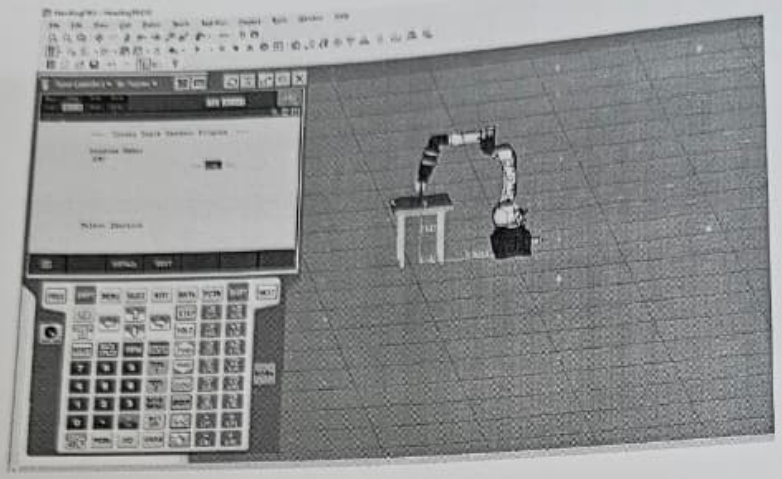




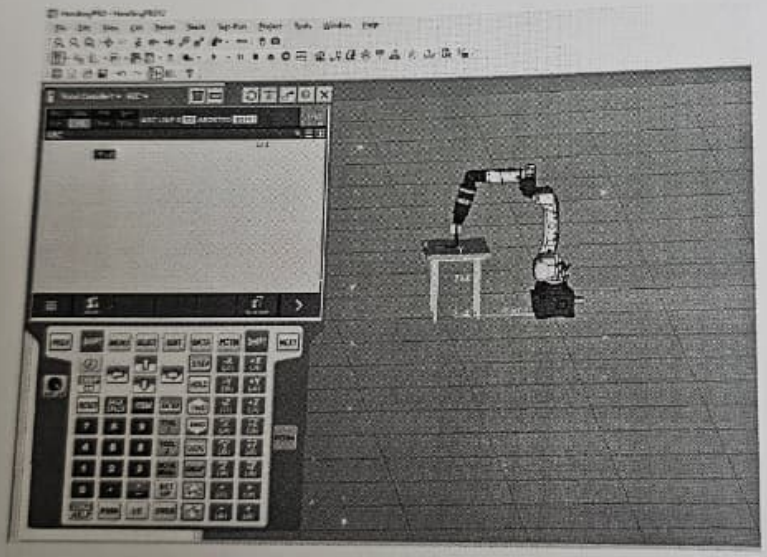
Click on the teach pendent



and enter

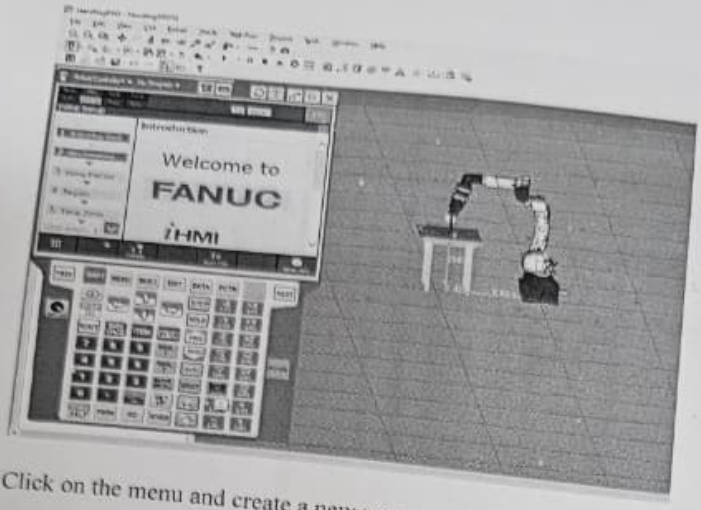


Enter the program here

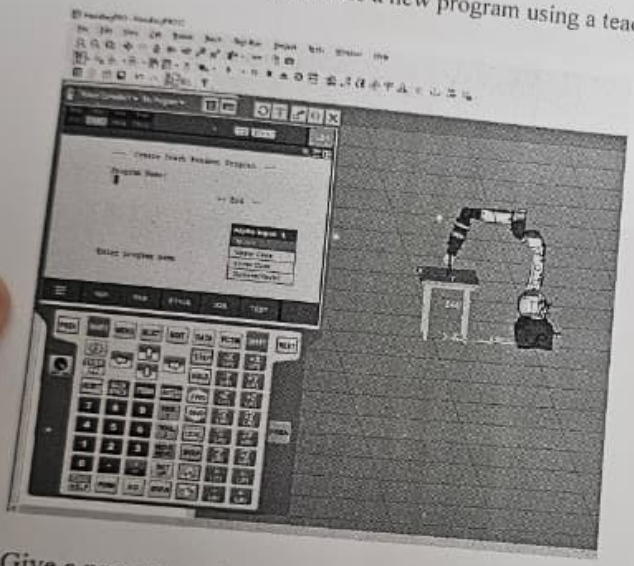


Inference:

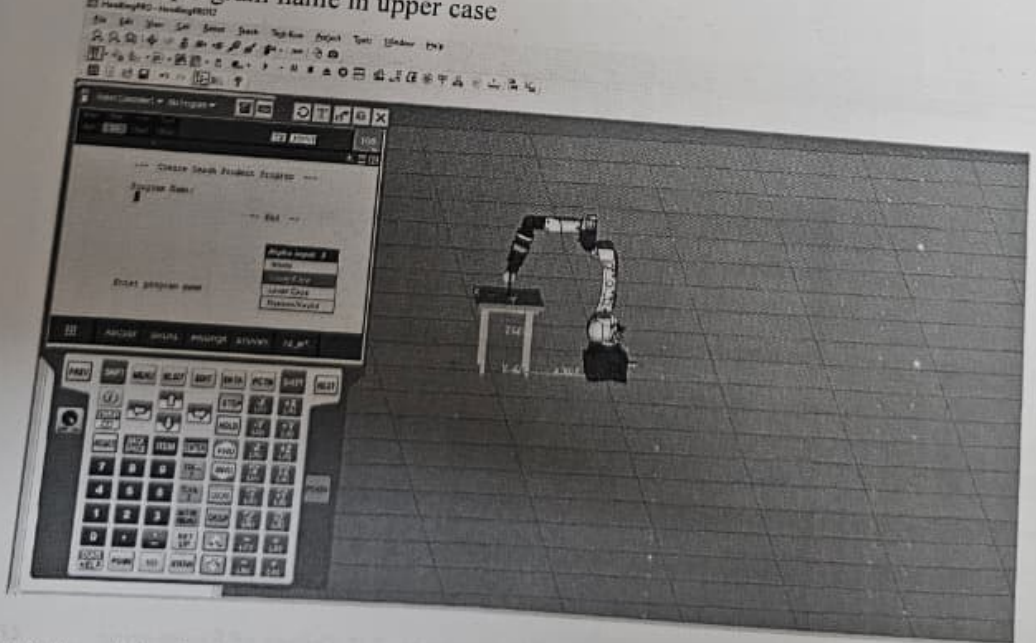
Particulars	Marks awarded	Marks obtained
1. Write-up of program	10	
2. Execution	5	
3. Results and Inference	2	
4. Viva	3	
Total	20	
Signature with Date		



Click on the menu and create a new program using a teach pendent

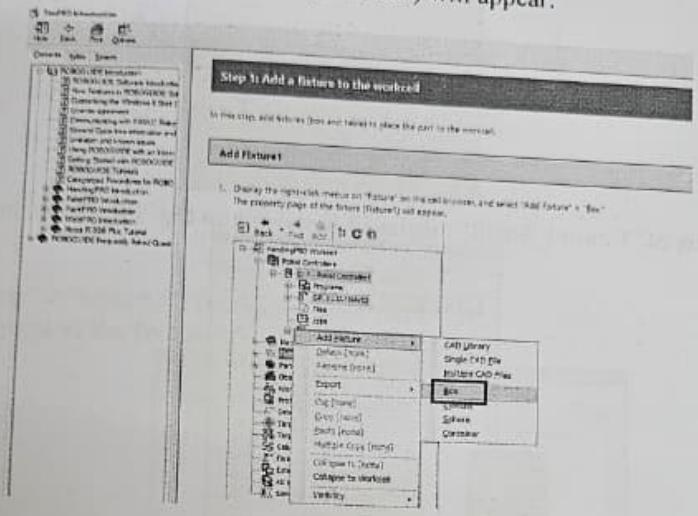


Give a program name in upper case

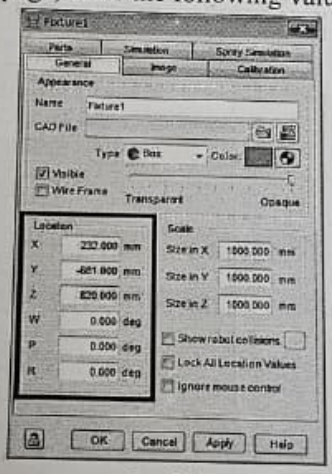


Roboguide: Pick and place of the work part using a Virtual Robot

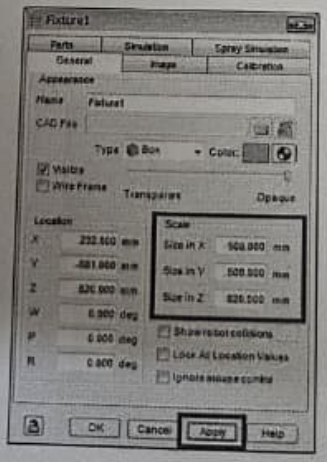
1. Display the right-click menus on "Fixture" on the cell browser, and select "Add Fixture" > "Box." The property page of the fixture (Fixture1) will appear.



2. On the "General" tab of the property page, enter the following values in "Location."



3. On the "General" tab of the property page, enter the following values in "Scale", and click "Apply."



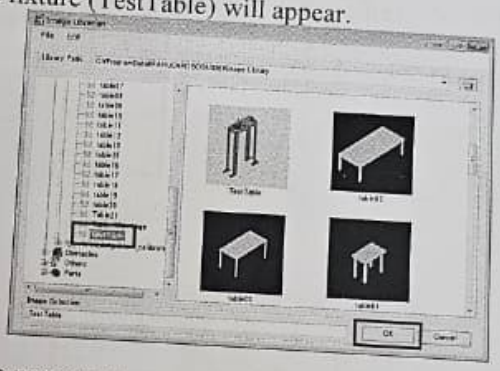
click "Apply."

ADD TESTTABLE

1. Display the right-click menus on "Fixture" on the cell browser, and select "Add Fixture" > "CAD Library."
   The CAD library screen will appear.



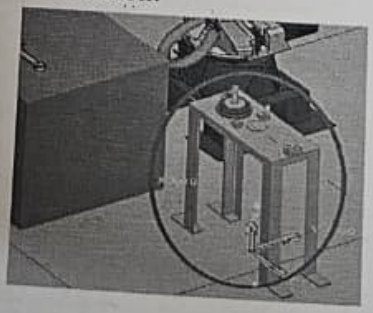
2. Select "Fixtures" > "table" > "TestTable", and click "OK."
   The property page of the fixture (TestTable) will appear.



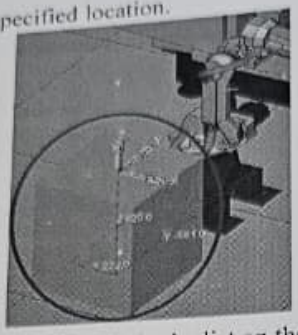
3. On the "General" tab of the property page, enter the following values in "Location", and click "Apply."



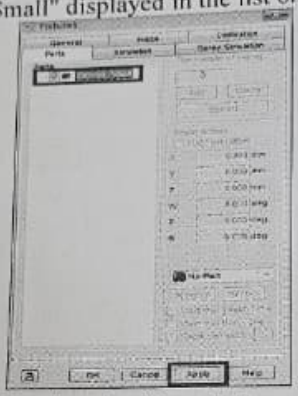
4. The fixture will appear at the specified location.



4. The fixture will appear at the specified location.



5. Check the box of "Conrod_Small" displayed in the list on the "Parts" tab, and click "Apply."



6. Check the box of "Edit Part Offset" on the "Parts" tab, enter the following values, and click "OK."



7. The part will appear at the specified location.



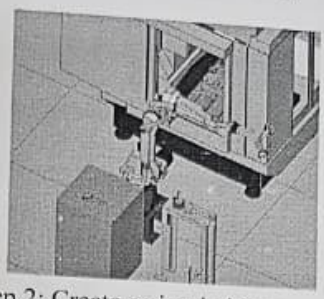
click "Apply."

and click

Roboguide status

When this step is completed, ROBOGUIDE is in the following state.

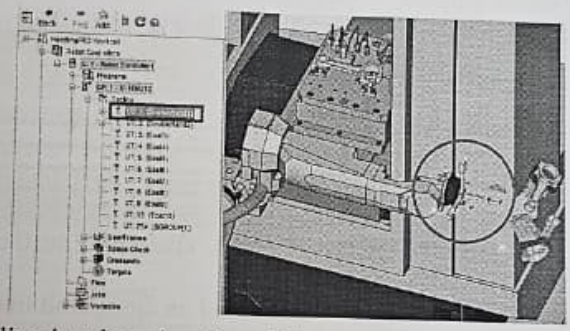
- The double hand motion is set.
- The program that opens and closes the double hand is created.
- The door motion is set.



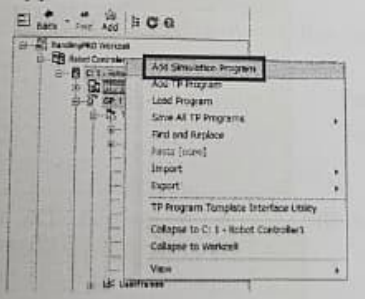
Step 2: Create a simulation program

Create a simulation program that grips the parts on the fixture

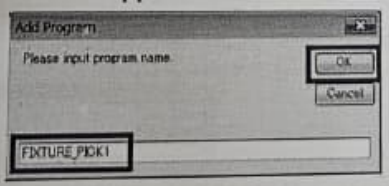
1. Click "Robot Controllers" > "C: 1 - Robot Controller1" > "GP: 1 - M-10iA/12" > "Tooling" > "UT: 1 (DoubleHand1)" on the cell browser.
   The tool frame of the tooling (DoubleHand1) is focused, and the triad will appear.



   Note:When the tooling is selected on the cell browser, the tool frame of the selected tooling can be used.
2. Select "Robot Controllers" > "C: 1 - Robot Controller1" > "Program" on the cell browser, and display the right-click menus. Then select "Add Simulation Program." The program creation screen will appear.



3. Enter "FIXTURE_PICK1" in the input box on the program creation screen, and click "OK." The simulation program edit screen will appear.

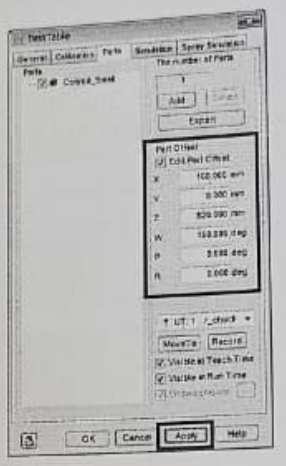


2

5. Check the box of "Conrod_Small" displayed in the list on the "Parts" tab, and click "Apply."



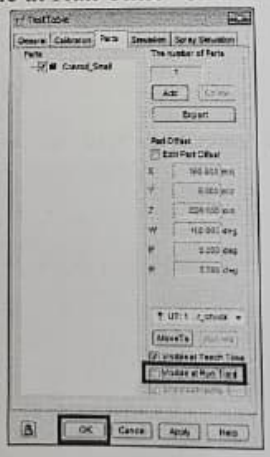
6. Check the box of "Edit Part Offset" on the "Parts" tab, enter the following values, and click "Apply."



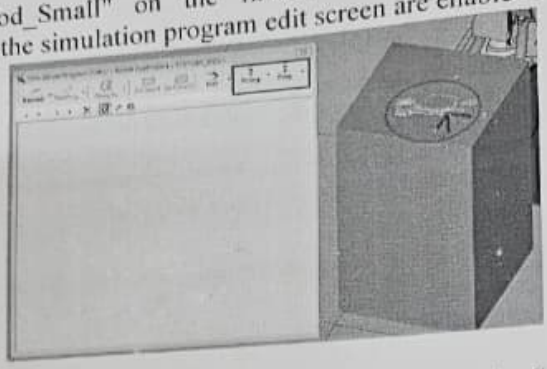
7. The part will appear at the specified location.



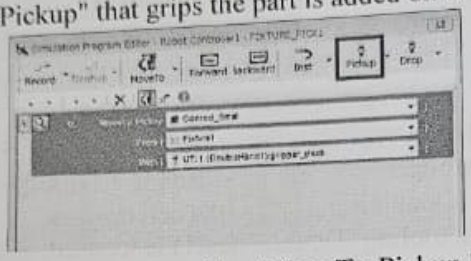
8. Remove the check mark from "Visible at Run Time" on the "Parts" tab, and click "OK."



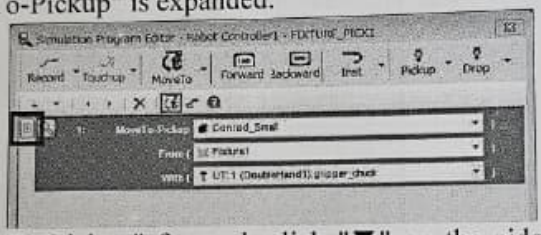
4. Click the part "Conrod_Small" on the fixture "Fixture1" on the graphic screen. "Pickup" and "Drop" on the simulation program edit screen are enabled.



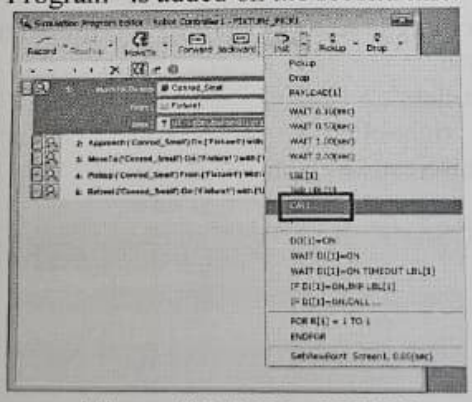
5. Click "Pickup."
   The program line "MoveTo-Pickup" that grips the part is added on the first line.



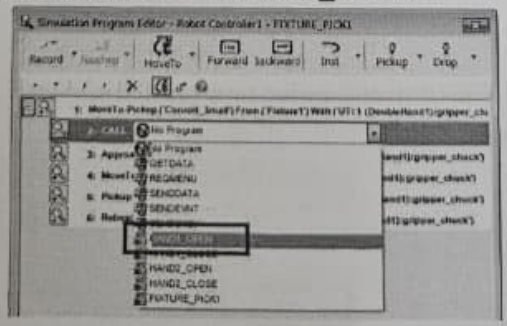
6. Click "+" on the left side of the program line "MoveTo-Pickup."
   The program line "MoveTo-Pickup" is expanded.



7. With the first line "MoveTo-Pickup" focused, click "▼" on the side of "Inst", and select "CALL..."
   The program line "CALL No Program" is added on the second line.

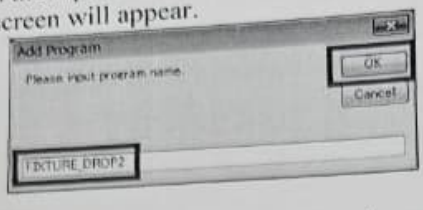


8. Change "No Program" on the second line to "HAND1_OPEN."

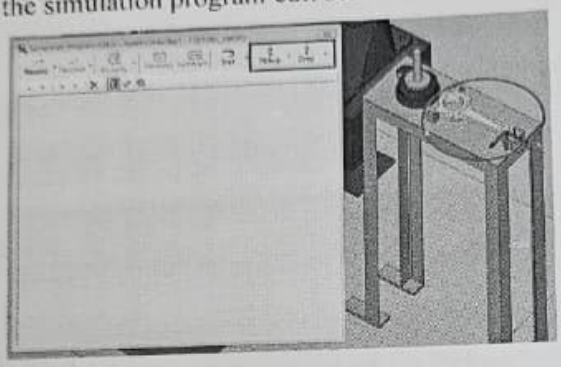


Note: Make the setting to call "HAND1_OPEN" to open DoubleHand1.

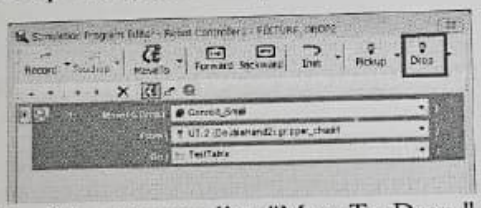
3. Enter "FIXTURE_DROP2" in the input box on the program creation screen, and click "OK."
   The simulation program edit screen will appear.



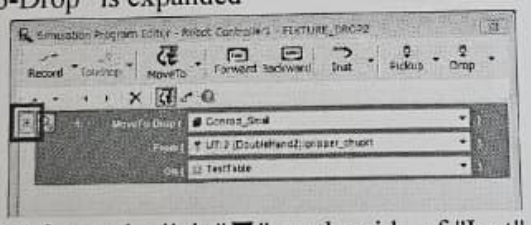
4. Click the part "Conrod_Small" on the fixture "TestTable" on the graphic screen.
   "Pickup" and "Drop" on the simulation program edit screen are enabled.



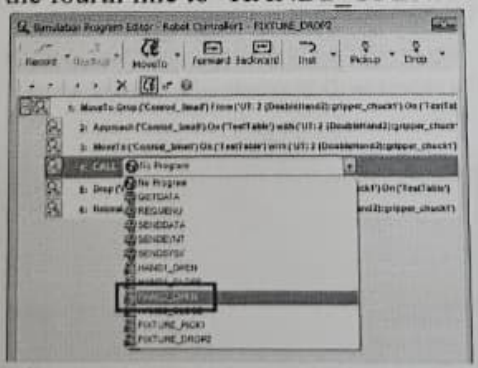
5. Click "Drop."
   The program line "MoveTo-Drop" that places the part is added on the first line.



6. Click "+" on the left side of the program line "MoveTo-Drop."
   The program line "MoveTo-Drop" is expanded

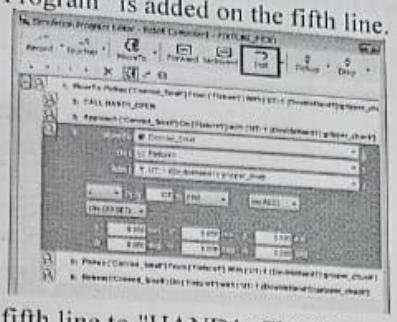


7. With the third line "MoveTo" focused, click "▼" on the side of "Inst", and select "CALL..."
   The program line "CALL No Program" is added on the fourth line.
   Change "No Program" on the fourth line to "HAND2_OPEN."

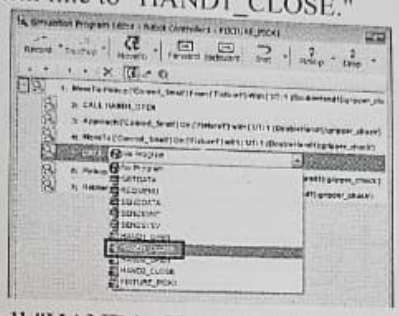


Note: Make the setting to call "HAND2_OPEN" to open DoubleHand2.

9. Click "Inst" with the fourth line "MoveTo" focused.
   The program line "CALL No Program" is added on the fifth line.



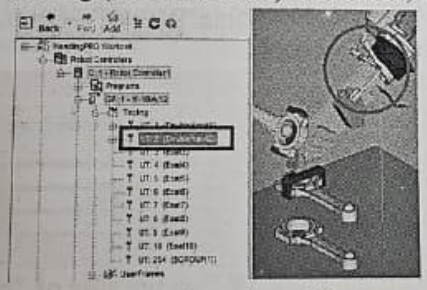
10. Change "No Program" on the fifth line to "HAND1_CLOSE."



Note: Make the setting to call "HAND1_CLOSE" to close DoubleHand1.

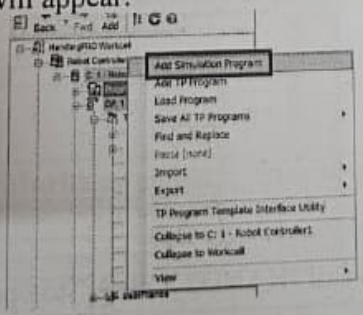
CREATE A SIMULATION PROGRAM THAT PLACES THE PART ON THE FIXTURE

1. Click "Robot Controllers" > "C: 1 - Robot Controller1" > "GP: 1 - M-10iA/12" > "Tooling" > "UT: 2 (DoubleHand2)" on the cell browser.
   The tool frame of the tooling (DoubleHand2) is focused, and the triad will appear.



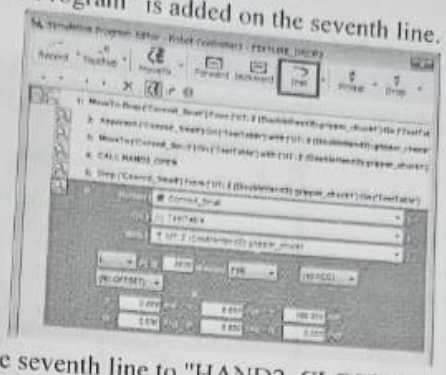
Note: When the tooling is selected on the cell browser, the tool frame of the selected tooling can be used.

2. Select "Robot Controllers" > "C: 1 - Robot Controller1" > "Program" on the cell browser, and display the right-click menus. Then select "Add Simulation Program."
   The program creation screen will appear.

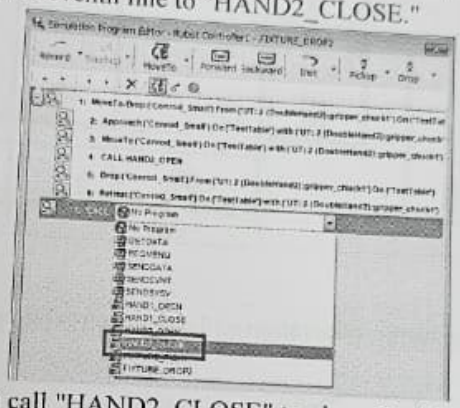


nd click "OK."

8. Click "Inst" with the sixth line "Retreat" focused.
   The program line "CALL No Program" is added on the seventh line.

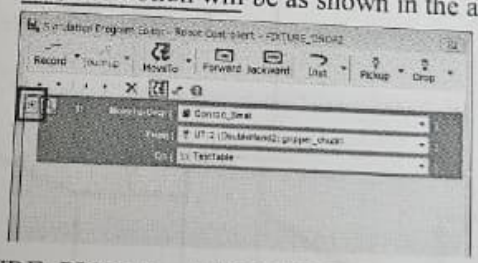


9. Change "No Program" on the seventh line to "HAND2_CLOSE."



Note:Make the setting to call "HAND2_CLOSE" to close DoubleHand2.

10. Select the programs "FIXTURE_PICK1" and "FIXTURE_DROP2" created in this step in turn on the cell browser, and click "Cycle Start ( )" on the toolbar.
    When each program is executed, the motion will be as shown in the animation below.



11. Select the programs "FIXTURE_PICK1" and "FIXTURE_DROP2" created in this step in turn on the cell browser, and click "Cycle Start ( )" on the toolbar.
    When each program is executed, the motion will be as shown in the animation below.

Particulars	Marks awarded	Marks obtained
1. Write-up of program	10	
2. Execution	5	
3. Results and Inference	2	
4. Viva	3	
Total	20	
Signature with Date		